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Plant growth-promoting rhizobacteria as a soil-based strategy to improve nutrient acquisition and drought tolerance in sugarcane

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Abstract

Drought is an increasingly important constraint to sustainable sugarcane production because prolonged water deficit can impair root development, nutrient acquisition, photosynthesis, biomass accumulation, sucrose accumulation, and ultimately cane and sugar yield. Plant growth-promoting rhizobacteria (PGPR) have emerged as a potential soil-based strategy for improving sugarcane performance under water-limited conditions through multiple direct and indirect mechanisms. This review critically evaluates the current evidence regarding PGPR-mediated drought mitigation in sugarcane, with special emphasis on nutrient acquisition, root-system development, phytohormonal regulation, ACC deaminase activity, biological nitrogen fixation, siderophore and extracellular polymeric substance production, antioxidant responses, and pathogen suppression. Available evidence suggests that PGPR can improve intermediate traits such as root growth, nutrient status, photosynthetic performance, water relations, and oxidative-stress regulation; however, the extent and consistency of these responses are highly variable among microbial strains, sugarcane genotypes, drought severity and duration, soil properties, climatic conditions, and management practices. Importantly, positive responses observed under laboratory, pot, and greenhouse conditions cannot be assumed to translate directly into stable yield benefits under field conditions. Severe or prolonged drought may restrict microbial survival, metabolic activity, root exudation, nutrient diffusion, and rhizosphere colonization, thereby reducing the effectiveness of inoculants. Evidence for several proposed mechanisms, including microbial hormone production, zinc, silicon, and sulphur mobilization, pathogen suppression under combined drought, and post-drought recovery, remains limited or indirect in sugarcane. Similarly, microbial consortia should not be assumed to be synergistic without appropriate comparative and factorial validation. Major priorities, therefore, include multi-location and multi-season field trials, standardized quantitative measurements of microbial persistence and plant responses, genotype- and soil-specific inoculant selection, longitudinal assessment across drought and recovery phases, and evaluation of economic feasibility and formulation stability. Overall, PGPR represents a promising



but context-dependent component of integrated drought management in sugarcane rather than a universally reliable standalone solution.

Keywords Sugarcane, Drought management, Plant growth-promoting rhizobacteria (PGPR), Rhizosphere, Nutrient acquisition, Microbial consortia

1 Introduction

Global climate change, together with the intensification of agricultural practices, has amplified both abiotic and biotic stressors, posing serious threats to global crop productivity and agroecosystem sustainability [72, 217, 249]. Among abiotic stresses, drought stress is widely recognized as one of the most severe constraints limiting global agricultural productivity that leads to significant reductions in crop productivity, particularly under prolonged water-deficit conditions coupled with high temperatures [195]. Climate projections indicate that drought-affected croplands may substantially expand by 2050, accompanied by significant declines in freshwater availability [79]. Drought stress severely affects plant physiology by impairing chlorophyll biosynthesis, reducing relative water content, limiting stomatal conductance, and restricting carbon assimilation, thereby leading to diminished biomass production and yield reductions in several cropping systems, including sugarcane [11, 108, 148]. At the molecular level, drought triggers complex regulatory responses involving differential expression of genes associated with abscisic acid (ABA) signaling, osmotic regulation, and reactive oxygen species (ROS) detoxification mechanisms [223, 241].

Sugarcane (*Saccharum* spp.) is a major industrial crop cultivated in more than 110 countries, with global production exceeding 2.0 billion metric tons in 2023, and Brazil and India together contributing more than half of the global output [80, 81]. Owing to its long growth duration, high biomass accumulation, and substantial water and nutrient requirements, sugarcane is particularly vulnerable to drought stress [84, 205]. Water-deficit conditions impair sugarcane root development, nutrient acquisition, stalk elongation, photosynthetic activity, and sucrose accumulation, often causing yield reductions of 50–70% depending on genotype, growth stage, and stress severity [138]. In addition to directly affecting plant physiology, prolonged drought alters rhizosphere microbial dynamics and nutrient cycling, thereby influencing plant-microbe interactions that may contribute to sugarcane drought adaptation [58]. Therefore, improving sugarcane resilience under water-limited conditions requires strategies that support plant growth, nutrient acquisition, and beneficial rhizosphere functions.

The use of rhizospheric microorganisms to buffer plants against drought stress is an emerging strategy in agriculture [31, 155, 203]. Plant growth-promoting rhizobacteria (PGPR) are safe alternatives to chemical inputs and improve plant nutritional status and abiotic stress tolerance by reducing drought stress in different crops through improvements in root architecture, water and nutrient uptake, phytohormone production, modulation of stress-responsive signaling, osmotic adjustment, antioxidant defence, and restoration of soil microbial diversity. The potential of genera and species like *Actinomyces*, *Azospirillum brasilense*, *Bacillus safensis*, *Burkholderia phytofirmans*, *Glucanacetobacter*, and *Ochrobactrum pseudogrignonense* has been proved in minimizing drought stress in various crops [126, 141, 142]. In sugarcane, drought management has relied mainly on genetic improvement, irrigation scheduling, nutrient optimization, and biotechnological approaches, with rhizosphere-based strategies increasingly explored as

a complementary approach. However, most supporting evidence still comes from controlled-environment and small-scale studies; multi-site field trials, assessments across diverse genotypes, long-term inoculant persistence data, and mechanistic verification across environments all remain limited [108, 115, 142, 184]. Given sugarcane's global economic importance and rising drought risk under climate change, this review examines the mechanistic basis, consistency, and field scalability of PGPR-mediated drought mitigation in sugarcane to identify the conditions under which these approaches can deliver reproducible benefits.

This review therefore critically discusses the current evidence for PGPR-mediated drought mitigation in sugarcane, including nutrient acquisition, root development and physiological responses, phytohormonal control, microbial stress-alleviating mechanisms, and rhizosphere interactions. Special attention is given to differences among microbial strains and consortia, sugarcane genotypes, drought severity and duration, and experimental conditions. The review also critically addresses inconsistent responses across studies and the challenges of extrapolating results from controlled trials to field conditions. Finally, research priorities are identified to advance mechanistic understanding, inoculant persistence, and multi-environment validation so as to define the conditions under which PGPR can deliver repeatable benefits to drought-stressed sugarcane production systems.

2 Drought stress in sugarcane

Erratic rainfall patterns and prolonged dry spells, intensified by climate change, have increasingly exposed many agricultural regions to water-deficit conditions, severely impairing crop development and productivity worldwide [104, 117, 173]. Projections indicate that by 2050, more than half of the world's arable land may experience severe water scarcity, posing a significant threat to global agricultural sustainability and crop production [112, 194]. The sensitivity of sugarcane to drought largely depends on the severity and duration of stress as well as the genotype involved. Drought adversely affects several physiological traits, including plant height, leaf area, photosynthetic activity, and nitrogen uptake, ultimately leading to significant reductions in sugar yield [20, 89]. To mitigate these challenges, sugarcane growers are increasingly adopting integrated management strategies, including drought-tolerant cultivars, efficient irrigation scheduling, biofertilizer application, and practices for conserving soil moisture [1, 209].

Under tropical and subtropical irrigated conditions, sugarcane has been reported to require approximately 20 million liters of water per hectare during a complete crop cycle; however, actual water requirements vary considerably depending on environmental conditions, crop duration, evapotranspiration demand, and management practices [199]. Exposure to drought during critical developmental stages, particularly the tillering and grand growth phases, severely limits sugarcane growth and productivity by inducing pronounced morphological changes, including increased tiller mortality, reduced cane height, shortened internodal length, decreased cane weight, lower millable cane population, reduced juice yield, and ultimately significant yield losses [28]. In addition, drought stress disrupts key physiological processes, such as photosynthesis, leaf ion balance, transpiration rate, cellular turgor, and assimilate partitioning, thereby limiting biomass accumulation and overall productivity [90, 242]. Consequently, water scarcity during peak growth stages not only restricts plant biomass but also reduces sucrose

accumulation, with yield losses reported to reach up to 60% under severe drought conditions [237]. In India, approximately 60–70% of sugarcane cultivation is carried out under rainfed or partially irrigated conditions, making the crop highly vulnerable to monsoon variability and drought stress [138, 209].

India and Brazil, two of the world's leading sugarcane-producing countries, share significant contributions towards the research and development of sugarcane [76]. Brazil is at the leading edge of research on the sugarcane microbiome, laying a strong foundation for climate-resilient approaches to sugarcane production systems [15, 212]. In the same vein, both countries have also made great strides in understanding microbial communities that are associated with sugarcane, biological nitrogen fixation, and microbial inoculants to facilitate sustainable sugarcane production [120, 136]. Therefore, ideas, insights, and innovations from India and Brazil can guide development of a globally applicable framework for improving drought resilience and sustainability through sugarcane growing globally. Under severe water-deficit conditions, yield reductions ranging from 30 to 60% have been reported, depending on genotype and irrigation management practices [2, 134]. Irregular rainfall patterns and declining groundwater availability significantly reduce cane yield and sugar recovery [9, 47, 96]. Similar drought-induced yield variability has also been documented in major sugarcane-producing countries due to frequent climatic aberrations [243, 254]. For instance, in Brazil, drought stress has been shown to significantly reduce cane yield and several yield-related parameters under prolonged water-deficit conditions [226]. Likewise, rainfed sugarcane cultivation in Thailand frequently encounters water deficits that substantially limit plant growth, biomass accumulation, and yield components under drought conditions [214].

3 Role of PGPRs in drought stress in sugarcane

The application of PGPR presents a sustainable and efficient alternative to mitigate the impacts of drought in sugarcane [68, 160]. These beneficial microbes support plant health through multiple mechanisms, including biological nitrogen fixation, mineral solubilization, and the production of bioactive compounds such as ammonia and hydrogen cyanide [25, 193, 225]. Inoculation with these rhizobacteria has been demonstrated to stimulate root system development, enhance antioxidant defense capacity, and upregulate photosynthesis-related gene expression, collectively improving plant physiological performance and drought resilience under water-limited conditions. In sugarcane, diverse rhizosphere-associated microbial genera, including *Arthrobacter*, *Microbacterium*, *Streptomyces*, *Bacillus*, and some strains of *Pseudomonas*, have been reported to mitigate drought stress by producing phytohormones and stress-alleviating metabolites that improve plant water status and physiological resilience [159, 162]. Other genera like *Azospirillum*, *Azotobacter*, *Burkholderia*, *Gluconacetobacter*, *Herbaspirillum*, and *Rhizobium* have also shown considerable potential to promote sugarcane growth, particularly under water-deficit conditions [67, 69]. Under field conditions, PGPR such as *Azospirillum brasilense*, *Bacillus subtilis*, and *Pseudomonas spp.*, applied individually or as microbial consortia, have been reported to enhance nutrient-use efficiency and improve sugarcane productivity through mechanisms including mineral solubilization, phytohormone production, stimulation of root proliferation, and enhanced rhizosphere activity that collectively promote plant growth and nutrient acquisition [41, 82]. However, studies comparing several microbial taxa in analogous environmental conditions

are few and far between, so no known universally better inoculants can be found. Future investigations should focus on comparing the candidate PGPR strains and also establishing multi-location field validations and genotype-specific assessments to confirm which of the microbial groups can reliably contribute to agronomic improvements under drought-prone sugarcane production systems.

The positive impact of PGPR inoculation on sugarcane performance has been studied earlier. For instance, Rosa, Galindo [177] reported that PGPR application reduced the requirement for phosphate fertilizers while simultaneously improving cane yield and technological quality parameters. Similarly, Tayade, Geetha [211] observed increased soil biological activity, enhanced rhizosphere functioning, and improved ratoon cane productivity following bio-intensive microbial modulation under tropical Indian conditions. Although many reports have been published on PGPR improving the performance of sugarcane under drought, it is difficult to directly quantify the PGPR effect in different experiments due to large differences in microbial strains, sugarcane genotypes, drought intensity, experimental duration, and environmental conditions. Thus, the reported results are significantly less likely to be consistent from study to study. Quantitative results on specific instances demonstrate that PGPR inoculation resulted in better root biomass, chlorophyll content, nutrient uptake, relative water content, and cane yield; however, the extent of the changes is varied among experimental settings [62]. Therefore, further research should offer similar information on effect magnitude, yield response, physiological factors, levels of microbial colonization, and environmental parameters. Standardizing these characteristics would enable future meta-analyses and help identify microbial strains that provide consistent gains in drought resistance across various sugarcane production systems.

The severity, length, and timing of drought stress greatly influence the effectiveness of PGPR in mitigating drought. The duration of drought is also important for plant and microbial responses. Short-term water deficits can induce adaptive physiological responses that may be partially mitigated by hormonal control and osmotic adjustment mediated by PGPR. Instead, persistent drought can cause lasting declines in photosynthesis, nitrogen uptake, and microbial activity. Previous studies have demonstrated that prolonged drought stress triggers the accumulation of phytohormones and osmolytes, leading to adaptive modifications in root architecture and enhanced stress tolerance in plants [245]. However, under severe and prolonged drought circumstances, reduced soil moisture and root exudates substantially impaired the microbial survival, root colonization, and metabolic activity of microorganisms [124, 143, 229]. The plant developmental stage affects drought responses and the efficiency of PGPR inoculation. Limitations of water during germination and the early developmental stages of sugarcane can reduce bud sprouting, tillering, root establishment, and canopy expansion. During the grand growth stage, it mainly limits stalk elongation, biomass accumulation, and nutrient uptake. However, harvest stages might affect sucrose accumulation, cane quality, and eventual yield [23, 103].

4 Factors influencing PGPR effectiveness under drought stress in sugarcane

Studies have shown that PGPR can improve the performance of sugarcane under drought conditions, although the efficacy of PGPR is strongly impacted by environmental, agronomic, and host factors. This suggests that drought remediation produced

by PGPR should not be considered equivalent to the same response in all production systems. However, the intensity and length of positive benefits are highly dependent on the severity of drought, crop growth stage, genotype, climatic circumstances, irrigation regime, and soil qualities. The efficiency of PGPR usually relies on the intensity of drought. Under moderate water deficiency, microbial inoculation stimulates root growth, nutrient uptake, osmotic adjustment, and antioxidant activity and hence enhances the performance of plants [10, 19].

The establishment and effectiveness of PGPR may vary with growth stages, since plant physiological requirements and root exudation patterns change during crop development. PGPR response variants have variability in genetics. Sugarcane cultivars differ significantly in root system architecture, rooting depth, water-use efficiency, osmotic adjustment capability, and drought tolerance potential. Root cultivars with deeper roots and drought-adapted cultivars with larger, deeper moisture reservoirs in the soil are more receptive to beneficial rhizosphere microorganisms, bettering root–microbe interactions in many circumstances. Thus, the efficacy of an inoculant in one genotype may not be easily transferred to other cultivars with distinct physiological properties [27].

The efficacy of PGPR also depends largely on environmental conditions. Sugarcane areas in the tropics usually have higher temperatures and longer growth seasons than subtropical areas with somewhat low winter temperatures. Such climatic changes are crucial in controlling microbial activity, processes of plant root growth, nitrogen cycling, and plant physiological responses to drought [57, 216]. Thus, PGPR strains acclimated to an agro-ecological location may show distinct climatic performances in comparison to another. Irrigation management can also affect the persistence and efficacy of imported microbes, as well as in rainfed systems, variations in soil moisture and the frequency of drought episodes may greatly influence microbial population dynamics and rhizosphere activity. However, supplementary watering may favor microbial survival and prolong positive plant-microbe interactions [10, 19, 144]. Therefore, irrigation regime has been emphasized as an important factor to be considered in field-induced inoculant performance.

Soil texture and physicochemical parameters also contribute to PGPR-induced drought mitigation. Sandy soils are inclined to moisture changes, which have been proposed to impair microbial colonization and survival. Sandy soils have a poorer water-holding capacity and limit nutrient retention in soil. Clay and loam soils, on the other hand, hold moisture and nutrients and tend to provide far more suitable conditions to maintain and exploit bacteria. Soil pH, organic matter content, and the indigenous microbial communities also affect the establishment and performance of introduced PGPR strains [55, 143]. These findings collectively indicate that the successful use of PGPR in sugarcane farming has to consider drought intensity, crop growth stage, genotype, meteorological conditions, irrigation techniques, soil type, etc. Further investigations are required in future to explore multi-location and multi-season field studies under diverse agro-ecosystem-based regimes to find out effective microbial strains and to optimize inoculant proposals for specific systems of sugarcane growth.

5 Rhizosphere as a determinant of PGPR efficacy under drought

The efficiency of PGPR under drought is majorly dependent on the rhizosphere. The dynamic interface between rhizosphere and microbes involves interaction between roots, imported and indigenous microorganisms, and soil physicochemical parameters. Drought simultaneously modifies root development and exudation, alters native microbial populations, lowers water-mediated nutrient transfer, and modifies soil pore-water distribution [99, 143]. Thus, the success of a PGPR strain is not only related to its functional features but also to its capacity to establish and maintain itself in this drought-modified environment.

Water deficiency alters root architecture, membrane permeability, and carbon allocation, which influences the quality and amount of root exudates (sugars, amino acids, organic acids, and phenolics) [88, 99]. These changes alter the carbon supplies accessible to the rhizosphere microorganisms, thus affecting the patterns of microbial recruitment. This phenomenon means that PGPR strains, depending on specific substrates released by the roots, may behave differently under drought, whereas strains already adapted to the changed resource environment may be favored [124, 220, 245]. However, there is as yet little direct sugarcane data tying specific drought-induced alterations in exudates to PGPR formation, unlike other plant systems. Changes in carbon supply due to drought affect not just the inoculated strain but normal bacterial and fungal populations. Inoculation thus happens in a pre-existing population of microbes rather than in an empty niche, which may help explain why beneficial *in vitro* PGPR performance typically does not translate to soil settings [125, 163, 239].

6 Soil physicochemical properties as determinants of PGPR performance

Soil physicochemical properties provide the physical and chemical framework within which these plant–microbe interactions occur. Soil pH affects the solubility of nutrients, the growth of microorganisms, enzyme activity, and the chemical forms of nutrients available to microorganisms for mobilization. pH affects the fixation of phosphorus, and thus the efficiency of organic acid-mediated phosphate solubilization by PGPR can be modified [105, 200]. Thus, a strain with high activity in solubilizing phosphate in one soil might exert a minor effect in another soil with a different pH and phosphorus chemistry. Soil texture and the amount of clay present are also important because they control the distribution of pore sizes, the amount of water retained, aeration, diffusion of nutrients, and the spatial distribution of the microorganisms. Conversely, clay-rich soils may offer increased water retention but may also change nutrient adsorption and microbial accessibility [44].

Soil organic matter supplies carbon substrates to promote microbial development and may influence the persistence of introduced inoculants under water-limited environments. However, availability of organic carbon does not necessarily guarantee effective establishment of PGPR, as introduced strains have to fight with indigenous bacteria for these resources. Similarly, the clay mineralogy can affect the potassium availability and nutrient-buffering capacity and hence modulate the magnitude of responses to potassium-solubilizing PGPR [6, 163]. These interactions also reveal why a strain selected for high nutrient-solubilization activity under laboratory conditions can operate differently when introduced into soils with variable pH, organic matter, texture, mineralogy, and nutrient status.

7 Integrated effects of drought, soil properties, and PGPR

The combined effects of drought and soil properties, along with PGPR, can generate substantial variation in PGPR performance across sugarcane agroecosystems. For example, severe drought may simultaneously reduce root activity and alter exudate availability, restrict microbial movement, decrease nutrient diffusion, alter the native microbiome, and increase competition for limited carbon resources [13, 99, 109]. Under such conditions, a PGPR strain may retain a desirable functional trait in principle but fail to express that trait at a level sufficient to influence plant performance [150, 164]. Under moderate water constraint, by contrast, greater root activity and rhizosphere connectivity may permit more effective microbial establishment and nutrient mobilization. This explains why PGPR responses are often stronger and more reproducible under variable drought conditions and their nutrient-solubilizing efficiency can be reduced by decreasing microbial growth, organic acid production, root colonization, and nutrient diffusion [190].

Therefore, PGPR efficacy should be evaluated as an emergent property of the plant–microbe–soil system rather than as an intrinsic characteristic of the inoculant. Future sugarcane experiments should simultaneously characterize root exudation, indigenous microbial community structure, inoculant abundance and persistence, soil moisture, pH, organic matter, texture/clay content, nutrient availability, and plant responses. Comparisons across contrasting soils and sugarcane genotypes would be particularly valuable for identifying the conditions under which a particular PGPR strain can successfully establish and provide reproducible benefits [44, 199]. This systems-level approach would also help distinguish true microbial inefficacy from failure of establishment caused by an unsuitable rhizosphere environment.

8 Beneficial activities of PGPR

Plant growth-promoting rhizobacteria (PGPR) alleviate drought-induced physiological, biochemical, and molecular constraints in sugarcane such as stomatal dysregulation, reduced photosynthetic efficiency, hormonal imbalance, nutrient uptake limitations, and oxidative stress by modulating plant hormonal signaling, enhancing root system architecture, improving nutrient availability, and stabilizing the rhizosphere environment. In the rhizosphere, plant-microbe interactions are effective in aiding crop development and mitigating the negative effects of drought on crops through nitrogen fixation, nutrient acquisition by antimicrobial metabolites, phosphorus solubilization, binding of microbial biofilms to plant roots, and induction of phytohormone production (Fig. 1) [12, 218, 221]. Under water-limited conditions, sugarcane experiences reduced cell turgor, impaired assimilate partitioning, and disruptions in hormone homeostasis. PGPR intervenes at these physiological bottlenecks by fine-tuning stress-responsive pathways, enabling sugarcane to maintain growth-defense balance. Rather than acting through isolated traits, PGPR function as integrated biological modulators that link drought physiology with microbial-mediated stress alleviation. These effects are broadly categorized into direct and indirect mechanisms, which collectively enhance drought resilience.

8.1 Direct mechanisms of PGPR in enhancing sugarcane drought tolerance

PGPR directly influence plant development by producing bioactive compounds and facilitating nutrient acquisition in the rhizosphere. Key mechanisms include the

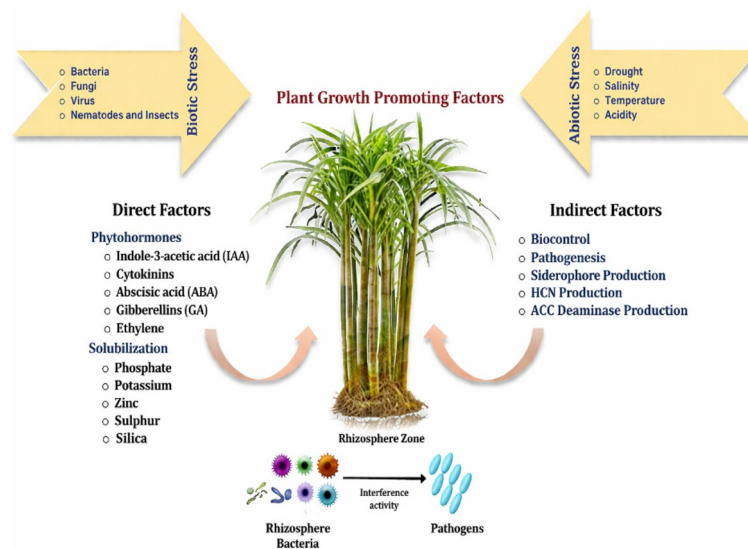


Fig. 1 Role of plant growth promoting rhizobacteria against biotic and abiotic factor through direct and indirect methods

synthesis of phytohormones such as abscisic acid (ABA), indole-3-acetic acid (IAA), gibberellic acid (GA), and cytokinins, all of which regulate plant physiological responses under stress conditions [108, 223]. They also enhance nutrient availability by solubilizing phosphorus, potassium, sulfur, and zinc and by producing siderophores that improve iron uptake under deficient conditions [95, 158].

8.1.1 Phytohormone-mediated stress mitigation

PGPR plays a central role in modulating plant hormones, particularly abscisic acid (ABA), indole-3-acetic acid (IAA), gibberellins (GA), and cytokinins (CK) as key regulators of water stress responses, helping balance growth and stress responses and coordinating root development, water conservation, and stress signaling Punpee [166]. However, how plants respond to hormones is based on concentration, and most plant hormones have optimum levels that often differ from those that induce growth inhibition or stress effects. For instance, moderate levels of indole-3-acetic acid (IAA) typically encourage root growth and lateral root production, whereas high IAA levels inhibit root growth and change root growth patterns [152, 179]. In addition, small-to-moderate accumulation of ABA controls stomatal regulation and drought adaptation, while prolonged or excessive ABA signaling impairs growth and carbon assimilation [10, 51]. Cytokinins, gibberellins, and ethylene have shown similar concentration-dependent effects. Despite these many reports, relatively few studies have quantified hormone concentrations directly in PGPR-inoculated sugarcane plants during drought. Thus, the effect of microbial hormone secretion, plant-level hormone accumulation, and drought tolerance is poorly understood. Furthermore, further studies that incorporate hormone quantification, microbial colonization estimation, and physiological validation are required to generate biologically plausible thresholds and to establish the role of PGPR-derived hormones to help adjust sugarcane to drought conditions. Although PGPR-associated production of these hormones has been reported, the concentrations achieved in sugarcane rhizospheres under field conditions remain largely unknown, making it difficult to

determine whether observed responses are directly attributable to microbial hormone production or to indirect regulation of plant hormonal pathways.

- Absciscic Acid

Absciscic acid (ABA) is a key drought-responsive hormone synthesized by plant roots, which controls the stomatal opening and closure and produces the first signal during plant stress [102]. ABA-mediated signaling reduces transpiration losses and enhances water-use efficiency while also promoting metabolic adjustment that prioritizes stress survival over vegetative growth [181, 252]. Exposure to moisture stress rapidly stimulates endogenous absciscic acid (ABA) biosynthesis in both roots and leaves by activating 9-cis-epoxycarotenoid dioxygenase (NCED), the key rate-limiting enzyme governing the ABA biosynthetic pathway (Fig. 2) [58, 227].

Molecularly, drought-related absciscic acid (ABA) accumulation activates PYR/PYL/RCAR receptors, inhibits PP2C phosphatases, and enhances SnRK2 kinase signaling pathways, leading to the expression of stress-responsive genes encoding LEA proteins, dehydrins, and ROS-scavenging enzymes [58, 247]. Transcriptomic analyses of sugarcane have shown upregulation of ABA-responsive genes under water deficit, providing indications that ABA-dependent signaling plays a conserved role in drought adaptation. It has been found that several plant growth-promoting rhizobacteria (PGPR), such as *Bacillus spp.*, *Azospirillum brasilense*, and *Pseudomonas spp.*, can secrete ABA in vitro [53, 194]. There is therefore scant direct evidence of the relationship between microbial ABA production and the activation of ABA-signaling pathways in drought-stressed sugarcane. While the majority of sugarcane studies concentrate on physiological responses, including water status, photosynthetic performance, antioxidant activity, and growth

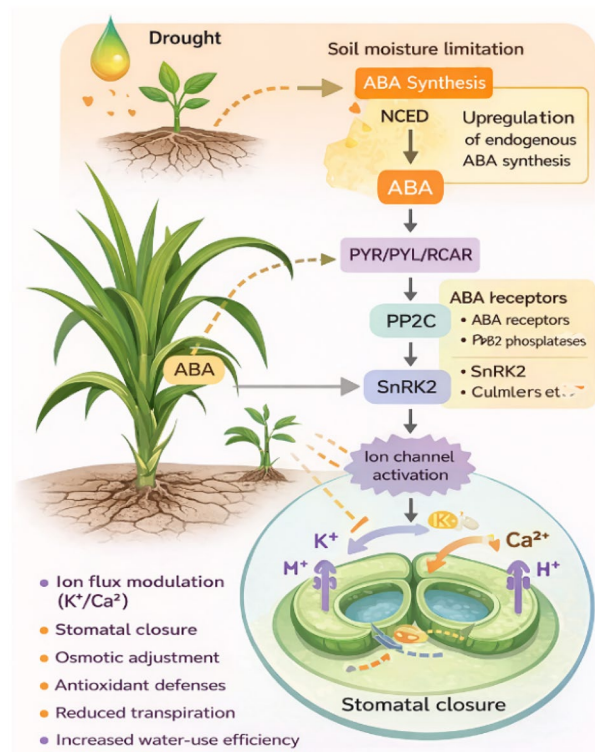


Fig. 2 Role of ABA in sugarcane drought stress response

promotion, the molecular characterization of PGPR-mediated ABA signaling pathways involved in this response has been relatively lacking. While drought-responsive ABA-related genes in sugarcane have been reported, few studies have explored microbial inoculation in combination with the dynamics of ABA and associated gene expression at the same scale. Consequently, it remains unclear whether PGPR-induced drought tolerance in sugarcane is primarily mediated through direct production of microbial ABA, induction of endogenous ABA biosynthesis, modification of ABA sensitivity, or ABA-independent mechanisms. Addressing this knowledge gap will require integrated transcriptomic, proteomic, metabolomic, and functional genomic investigations under controlled and field conditions.

- Indole-3-Acetic Acid

Sugarcane, a highly water-demanding crop, relies strongly on auxin-mediated root system plasticity to sustain water uptake and physiological functions under drought stress [210]. Under water-limited conditions, optimal levels of IAA promote primary root elongation, lateral root formation, and root hair development, enabling plants to access deeper and less-depleted soil moisture reserves. In contrast, excessive auxin accumulation can suppress primary root growth while promoting lateral root proliferation, reflecting the dual and context-specific nature of auxin signaling [71, 251]. This adaptive response is further strengthened by IAA-producing rhizobacteria, which enhance nutrient acquisition and water-use efficiency and contribute to sustained stalk biomass and yield stability by modulating local auxin gradients in the rhizosphere [34, 187, 206]. Collectively, the integration of plant - and microbe-derived IAA signaling represents an important mechanism contributing to drought tolerance in sugarcane. *Rhizobium*, *Azospirillum*, *Azotobacter*, *Microbacterium*, *Mycobacterium*, *Pseudomonas*, *Stenotrophomonas*, *Sphingomonas*, and *Arthrobacter* strains are some of the most well-known IAA-producing rhizobacterial genera [7, 33]. At the molecular level, auxin signaling during drought involves regulation of the *AUX/IAA*, *GH3*, and *SAUR* gene families, which interact with abscisic acid (ABA) signaling pathways to regulate antioxidant defense and cell wall plasticity [181, 247]. The crosstalk between auxin and ABA signaling is increasingly recognized as a key regulatory module coordinating adaptive root remodeling and growth regulation under water-deficit conditions (Fig. 3). In sugarcane, drought-induced hormonal modulation has been associated with altered root development and the activation of stress-responsive genes [123, 174], further highlighting the integrative role of auxin signaling in drought adaptation.

While microbial indole-3-acetic acid (IAA) production has been suggested as a promising path to alleviate drought stress, quantitative data based on IAA in rhizosphere and plant tissues from PGPR-inoculated sugarcane are limited. Therefore, the ranges of concentrations at which IAA have a beneficial or a harmful effect in sugarcane are poorly characterized, and most current interpretations are based on evidence from model plants and other crop species instead of data from sugarcane-specific dose-response studies. This knowledge gap limits our understanding of the physiological relevance of microbial IAA production under drought conditions. IAA also has an extensive interaction with ethylene synthesis. Elevated auxin concentrations can stimulate the expression of ACC synthase, the key enzyme involved in the synthesis of 1-aminocyclopropane-1-carboxylate (ACC), the immediate precursor of ethylene [246]. This interaction

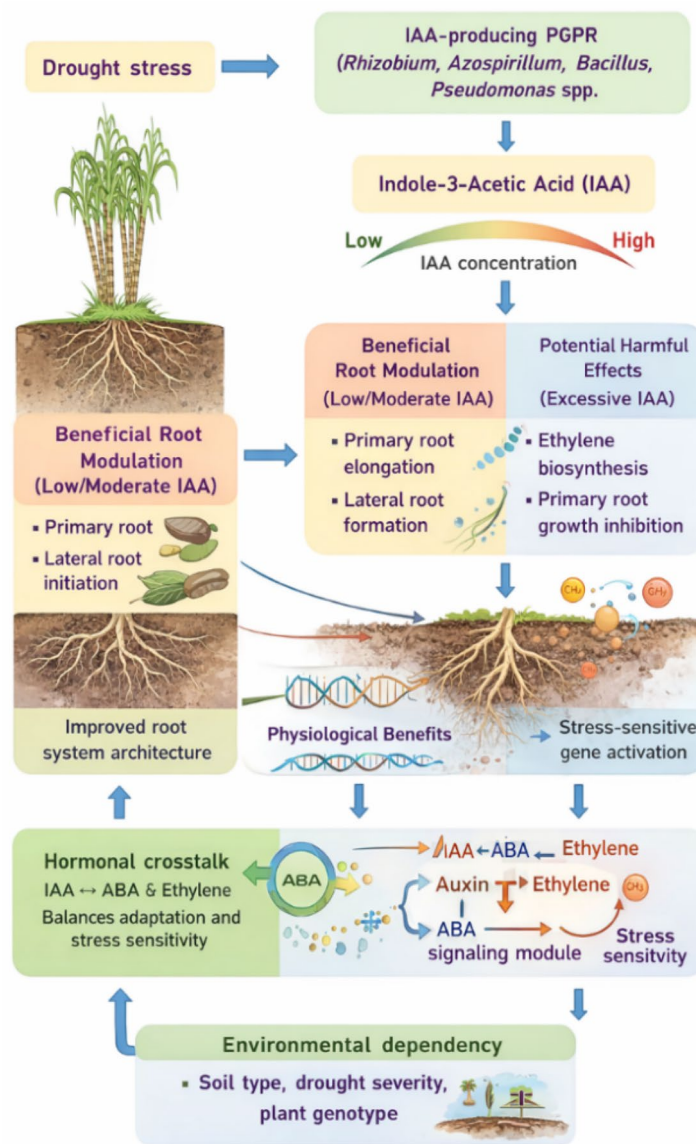


Fig. 3 Role of IAA in drought stress tolerance in sugarcane

highlights the importance of hormonal balance rather than absolute hormone production. In this context, ACC deaminase-producing PGPR may play a particularly important role by reducing ACC availability and limiting excessive ethylene accumulation, thereby helping maintain the beneficial effects of auxin-mediated root development during drought stress [10, 19, 91].

- Gibberellic Acids

Gibberellic acids (GAs) produced by PGPR regulate diverse plant developmental processes, including seed germination, stem elongation, leaf expansion, and reproductive development. The cellular concentration of bioactive GAs is tightly controlled through the coordinated action of biosynthetic enzymes, including GA20-oxidase (GA20ox) and GA3-oxidase (GA3ox), together with GA-deactivating enzymes such as GA2-oxidase

(GA2ox) (Fig. 4) [98, 244]. Gibberellin metabolism is closely interconnected with auxin signaling, as auxin-responsive transcription factors can regulate the expression of GA biosynthetic genes such as *GA20ox* and *GA3ox*, thereby modulating endogenous GA levels and coordinating cell elongation and growth responses under both normal and stress conditions [87, 178]. Under drought conditions, GA metabolism exhibits pronounced spatial and temporal variability rather than uniform downregulation. In many crops, including sugarcane, interactions between auxin and gibberellin signaling are mediated through complex hormone regulatory networks, where auxin-responsive transcription factors influence growth-related gene expression and may indirectly modulate GA biosynthesis pathways involved in stem elongation, leaf expansion, and developmental regulation [54, 102, 122, 123].

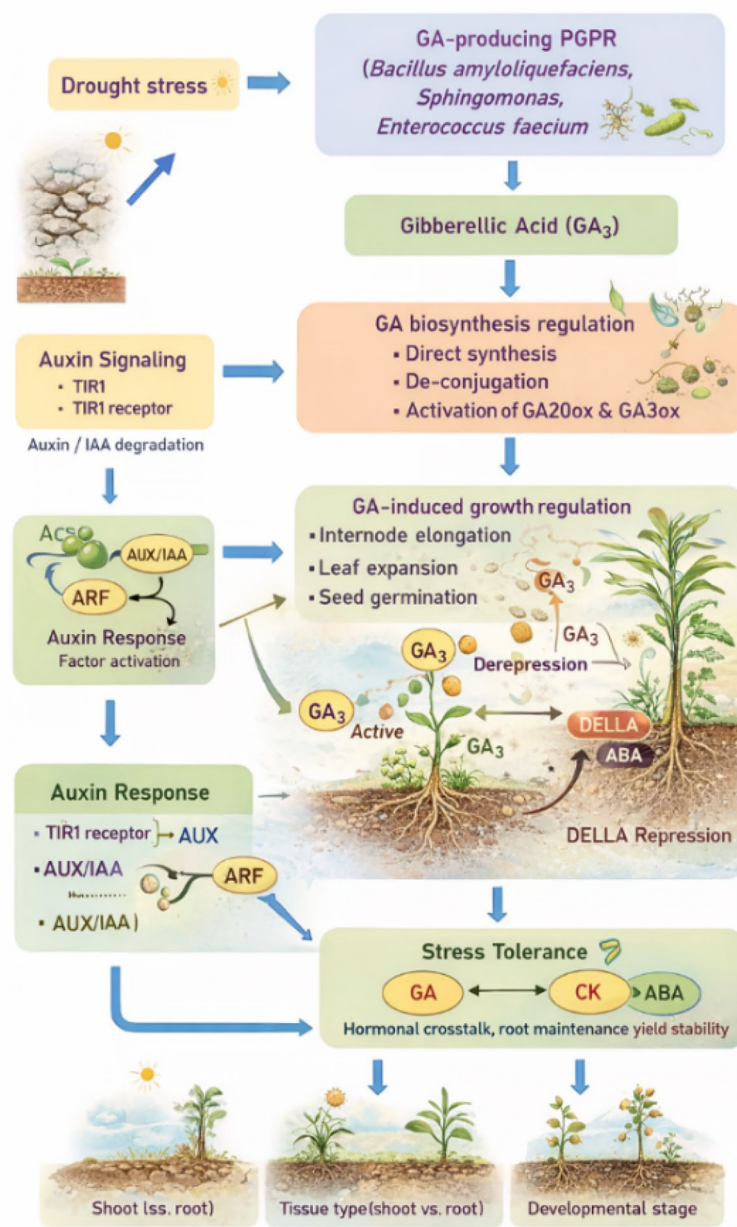


Fig. 4 Role of gibberellic acid and GA-producing PGPR in sugarcane drought stress tolerance

At the molecular level, drought frequently promotes the accumulation of DELLA proteins, which are negative regulators of GA signaling. Excessive accumulation of DELLA proteins under drought exposure leads to growth inhibition and stress tolerance [3, 54]. It has been suggested that the PGPR genera like *Bacillus*, *Azospirillum*, and *Pseudomonas* produce GA-like compounds; however, direct evidence showing microbial GA production was significantly involved in drought tolerance in sugarcane is still limited [19, 39]. Although most studies demonstrate growth and physiological performance improvements post-PGPR inoculation, they do not directly measure microbial GA production, plant GA concentrations, or DELLA-activated signaling responses. Thus, the available evidence is indicative of the involvement of microbial gibberellins as a potential addition to the multiple hormonal and physiological pathways, rather than a sole drought-mitigation system.

- Cytokinin

Cytokinins (CK) play a central role in regulating cell division, shoot growth, and leaf development [135, 166]. However, under drought stress, endogenous CK levels typically decline, shifting the growth balance toward enhanced root proliferation. Reduced cytokinin levels favor the development of deeper and more extensive root systems, enabling plants to access water from deeper soil layers. This adaptive shift is often accompanied by increased abscisic acid (ABA) accumulation, which improves stomatal regulation and enhances drought resilience [61, 119]. Experimental evidence indicates that manipulation of cytokinin levels can significantly influence plant drought responses. Several studies have shown that exogenous cytokinin application improves plant growth and stress tolerance under water-limited conditions in diverse plant species, including sugarcane [131], pineapple [16], *Arabidopsis thaliana* [207], Kentucky bluegrass [100], and ryegrass [232]. Under moderate water-deficit conditions (approximately 50% field capacity), cytokinin-enhanced plants maintained about 30–40% higher chlorophyll content and 40–50% greater green leaf area [145].

The quantitative estimation of PGPR-derived cytokinins against drought tolerance is scarce in sugarcane. One of the significant limitations of current research is that no studies currently provide simultaneous quantitative analysis of microbial cytokinin production, endogenous cytokinin concentrations, microbial colonization levels, and drought-response capacities in sugarcane. Thus, the role of PGPR-derived cytokinins in the adaptation to drought in plants is still unknown. While certain bacterial strains (such as *Bacillus*, *Azospirillum*, *Pseudomonas*, and *Paenibacillus*) have been identified as capable of producing cytokinin-like compounds in culture-based experiments. Studies by several researchers indicate that the favorable effects of PGPR for sugarcane in drought-stricken conditions may be due to more robust physiological processes in root growth, nutrient acquisition, osmotic adaptation, antioxidant and hormonal regulation pathways, than only the production of cytokinins (Fig. 5) [41, 86, 147].

9 Nutrient solubilization by PGPR under drought stress in sugarcane

Drought stress not only reduces water availability but also significantly alters soil physicochemical and biological processes, thereby restricting nutrient mobility and uptake. Reduced soil moisture lowers the diffusion rates of relatively immobile nutrients such as phosphorus (P) and zinc (Zn) while also limiting mass flow-driven transport of mobile

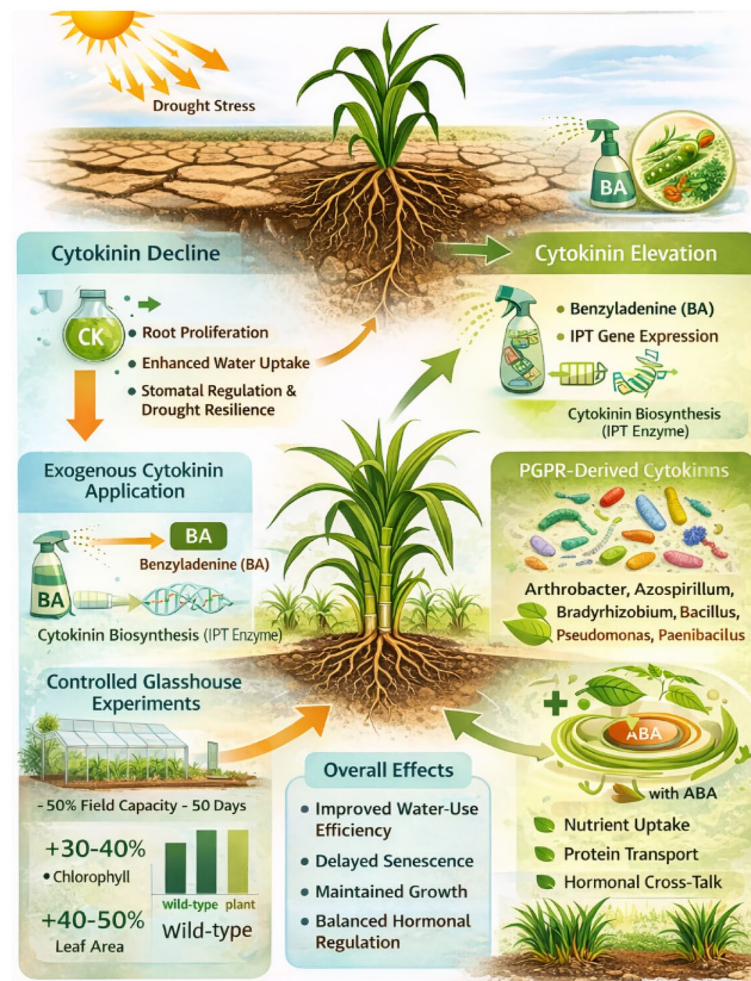


Fig. 5 Role of Cytokinins in sugarcane growth and stress adaptation under water-Limited condition

nutrients, including nitrate and potassium (K) [101, 189]. Consequently, nutrient deficiency under drought arises not only from reduced nutrient mobility but also from the disruption of soil–root–microbe interactions that normally sustain nutrient cycling processes. PGPR can mitigate these limitations by maintaining nutrient mobilization processes in the rhizosphere despite reduced soil moisture. When diffusion-limited mineral acquisition becomes critical during drought, localized microbial activity near root surfaces can partially compensate for restricted nutrient transport [222]. In addition to nutrient solubilization, PGPR also improve nutrient acquisition by modifying root system architecture and stabilizing the rhizosphere environment [45]. Furthermore, microbial production of extracellular polysaccharides (EPS) improves soil aggregation and water retention around roots, creating protective microenvironments that maintain nutrient exchange and enzymatic activity under water-deficit conditions [183]. Numerous studies have reported enhancing nutrient acquisition under drought conditions; however, direct evidence demonstrating sustained nutrient-solubilization activity under severe water deficit remains limited (Table 1). Most nutrient-solubilization mechanisms have been characterized under laboratory or well-watered conditions, whereas their effectiveness under prolonged drought is less well understood. Organic acid production,

Table 1 Rhizosphere-mediated mineral solubilization and associated plant growth-promoting effects in sugarcane under water deficit conditions

Rhizobacteria	Mineral solubilized	Mechanism	Impact on crop	Reference
<i>Bacillus subtilis</i>	Phosphorus, Potassium	Production of organic acids (e.g., citric and gluconic acids)	Enhanced nutrient uptake and root growth	[234]
<i>Pseudomonas fluorescens</i>	Phosphorus, Iron	Siderophore production and organic acid secretion	Improved drought tolerance and biomass	[130]
<i>Azospirillum brasilense</i>	Phosphorus	Acidification and chelation of insoluble phosphates	Increased root elongation and sugar yield	[21]
<i>Rhizobium leguminosarum</i>	Phosphorus, Zinc	Organic acid production and enzymatic mineralization	Boosted growth under water deficit	[245]
<i>Arthrobacter</i> spp.	Phosphorus, Calcium	Secretion of acid phosphatases and chelating agents	Enhanced stress resilience and growth	[107]
<i>Burkholderia cepacia</i>	Phosphorus, Silica	Organic acid-mediated mineral dissolution	Improved plant vigor and stress tolerance	[56]
<i>Acinetobacter</i> spp.	Zinc, Phosphorus	Enzymatic hydrolysis and acidification	Better growth and sugar accumulation	[116]

enzymatic activity, microbial growth, and rhizosphere colonization are all sensitive to soil moisture availability (Fig. 6).

Thus, the performance of phosphate, potassium, zinc, sulphur, and silicon mobilization during droughts may potentially vary significantly from that of the best-performing conditions of moist conditions. It has been also observed that the efficiency of nutrient solubilization of PGPR is largely affected by soil physicochemical conditions. The soil pH determines the dominant forms of phosphorus fixation and affects the effectiveness of organic-acid-mediated solubilization [105, 200]. Similarly, soil texture will have an impact on water retention, nutrient diffusion, and microbial survival, and drought effects are commonly more severe in coarse-textured sandy soils than in fine-textured clay soils [44]. Carbon substrates of organic matter provide the platform for microbial growth and may dictate the persistence of an introduced inoculant in a water-limited ecosystem [64, 150]. In addition, clay mineralogy modifies potassium availability and nutrient buffering capacity, which, accordingly, alters the efficacy of microbial nutrient mobilization [6, 163]. These dynamics indicate that nutrient-solubilizing PGPR are unlikely to be consistently effective in all sugarcane-growing landscapes, underscoring why soil and climatic evaluation must be site-specific [150, 231].

- Phosphorus (P)

In sugarcane, adequate phosphorus supply is crucial for tillering, nutrient absorption, and sucrose synthesis [42, 123, 156]. A wide range of bacterial genera, including *Azotobacter*, *Bacillus*, *Pseudomonas*, *Rhizobium*, and *Enterobacter*, possess phosphate-solubilizing capabilities. However, the effectiveness of these microbes under drought

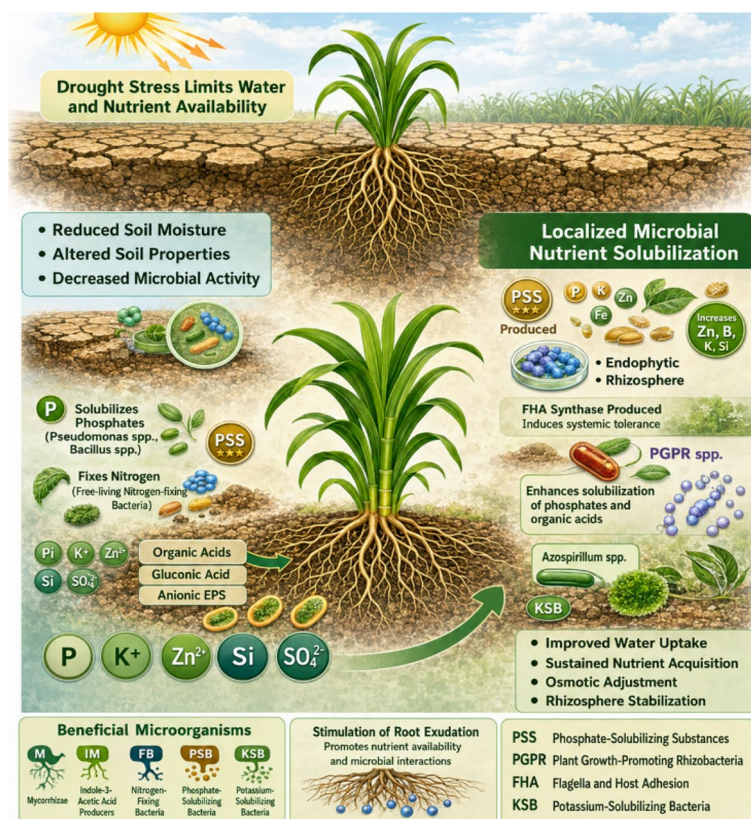


Fig. 6 Nutrient Solubilization by PGPR under drought stress in sugarcane

conditions in sugarcane is often strain-specific and influenced by environmental factors. While many studies observe increases in plant phosphorus concentrations after inoculation with PSB, the responses vary substantially and depend on the microbial strain, soil characteristics, phosphorus availability, drought severity, and sugarcane genotype [151, 175, 198]. In sugarcane, similar variability has been observed under different soil and management conditions [208]. In several moderate water-deficit experiments, these improvements in plant P status have been associated with 8–18% increases in cane yield and enhanced sucrose content when microbial inoculation was integrated with 75–100% of the recommended P fertilizer dose [177, 186].

Despite these benefits, the effectiveness of PSB is strongly influenced by drought severity and soil phosphorus availability. In moderate water deficits, microbial growth, organic acid production, and root colonization can still be promoted after PSB and thus increase phosphorus availability [29, 230]. But during extreme or prolonged drought, the metabolic activity of microflora, root growth, and nutrient diffusion can be significantly reduced, which reduces the efficiency of phosphate solubilization [38, 256]. Thus, PSB-mediated improvements are more stable under mild-to-moderate drought than under extreme water scarcity. Soil phosphorus status is also a significant determinant of the effectiveness of phosphate-solubilizing bacteria. Responses are primarily greatest in soils containing considerable amounts of insoluble phosphorus that can be mobilized, for example, by microbial activity.

- Potassium (K)

Potassium (K) is an essential macronutrient that plays a critical role in enzyme activation, osmotic regulation, stomatal movement, photosynthesis, and stress signaling in plants [43]. Potassium-solubilizing bacteria (KSB), specifically in the genera *Bacillus* and *Pseudomonas*, have been described as potential potassium mobilizers playing a crucial role in enhancing K bioavailability under water-limited conditions by creating organic acids and weathering minerals. The evidence base for K-solubilization-infused drought responses is largely derived from wheat, maize, and other cereals, where the inoculation has resulted in higher uptake of potassium, antioxidant potential, and stress tolerance [76, 133, 137]. In sugarcane, potassium is required for stomatal regulation, osmotic adjustment, carbohydrate translocation, and maintenance of photosynthetic activity, but little research is available to confirm directly the role of potassium-solubilizing PGPR in these processes under drought conditions. As a result, additional validation of such mechanisms identified in other crops, which apply here to sugarcane, must also be ensured. Although enhanced potassium nutrition may theoretically facilitate stomatal regulation and osmotic adjustment and improve plant water relations during drought conditions, their direct measurements remain scarce in sugarcane [128, 170, 199]. Collectively, these findings highlight that microbial potassium solubilization represents an important biological strategy for alleviating drought-induced nutrient limitations and enhancing crop resilience, particularly in water-scarce and nutrient-depleted soils.

- Zinc (Zn)

Zinc is essential for enzyme function, hormone biosynthesis, and plant metabolism and is especially crucial during stress responses. Under drought stress, reduced soil moisture negatively affects zinc mobility by restricting diffusion and mass flow toward roots,

thereby exacerbating Zn deficiency [143]. Several plant growth-promoting rhizobacteria (PGPR), including *Gluconacetobacter diazotrophicus*, *Bacillus subtilis*, *Enterobacter cloacae*, and *Pseudomonas aeruginosa*, have demonstrated zinc-solubilizing capabilities. In vitro studies have demonstrated that bacterial isolates have the potential to solubilize insoluble zinc compounds such as ZnO, ZnCO₃, or Zn₃(PO₄)₂ [172, 188]. However, effective solubility does not necessarily lead to an increased availability of Zn²⁺ in the rhizosphere, zinc uptake by roots, or improved plant functioning under dry conditions. Factors that affect the amount of solubilized zinc available for plant uptake include soil chemistry, microbial survival, root activity, and drought severity. To date, few studies have quantitatively assessed the microbial zinc-solubilizing activity, plant zinc accumulation, and physiological responses simultaneously under water-deficit circumstances in sugarcane [140, 219]. In comparison with other systems such as phosphorus solubilization, biological nitrogen fixation, or phytohormone-induced root growth, microbial zinc mobilization's contribution towards drought adaptation in sugarcane is relatively uninvestigated. Therefore, zinc solubilization should be treated up to now as a potentially advantageous but not well described mechanism to be experimentally confirmed. Zinc solubilizing bacteria have shown good potential in laboratory- and greenhouse-based experiments. However, the available evidence is not sufficient to conclude that the microbial zinc solubilization significantly enhances zinc uptake and drought tolerance in sugarcane for field applications. More current studies should connect microbial zinc mobilization with plant zinc nutrition, physiological responses, and yield performance under varied drought circumstances.

- Silicon (Si)

Silicon (Si) is not classified as an essential nutrient for most plants; however, it is widely recognized as a beneficial element, particularly in monocot crops such as rice, wheat, and sugarcane [74, 129]. Under drought stress, silicon improves water-use efficiency, structural reinforcement of cell walls, reduced transpiration, enhanced antioxidant defense, and stabilization of photosynthetic machinery [94, 257]. In soil, silicon is primarily present in insoluble forms such as insoluble silicate minerals, including feldspars, quartz, and clay-associated silicates, or as silica (SiO₂), whereas plants absorb silicon in the soluble form of monosilicic acid (H₄SiO₄) [129]. Several bacterial genera, including *Bacillus*, *Pseudomonas*, *Burkholderia*, and *Enterobacter*, have been reported to solubilize silicate minerals under laboratory conditions [133, 224]. However, there is no direct evidence for microbial silicate weathering and increased H₄SiO₄ availability in drought-stressed sugarcane rhizospheres [77, 110].

Under moderate drought conditions, microbial enhancement of Si availability has been associated with reduced transpiration rates, improved relative water content, and enhanced photosynthetic performance in several crops [94]. For example, inoculation with silicon-mobilizing *Pseudomonas fluorescens* has been shown to increase soluble Si availability and improve physiological drought tolerance in maize under controlled moisture stress [224]. As a high silicon-accumulating crop, sugarcane benefits from adequate Si nutrition through improved stalk strength, enhanced leaf erectness, and greater tolerance to water deficit and lodging stress [129]. In sugarcane production systems, soil or foliar silicon supplementation, often interacting with microbial activity, has also been linked to improved cane yield and sugar recovery, particularly in Si-deficient soils

[111]. The efficiency of silicon-based interventions is thus strongly dependent on the initial soil silicon supply. Positive responses are often more evident in silicon-deficient soils where the amount of plant-available silicon is low and where extra silicon inputs can significantly increase absorption. The estimation of agronomic value of the silicon-solubilizing microorganisms in sugarcane production systems should consider the soil mineralogy, weathering status, and existing silicon availability. However, there is a significant drawback of existing research is the lack of field studies that correlate microbial silicate weathering with soil monosilicic acid elevations, plant silicon accumulation, and drought-tolerance responses of sugarcane.

- Sulfur (S)

Drought stress reduces sulfate mobility and mass flow toward roots due to lower soil moisture, thereby limiting sulfur uptake and disrupting metabolic and antioxidant processes critical for stress tolerance [121]. Sulfur-mediated enhancement of glutathione, cysteine, and sulfated secondary metabolites improves ROS scavenging and stabilizes cellular membranes under water deficit [116]. Sulfur-solubilizing bacteria (SSB) notably certain strains of *Bacillus*, *Pseudomonas*, *Thiobacillus*, and mycorrhizal fungi can mineralize organic sulfur compounds into free sulfate ions (SO_4^{2-}), increasing plant-available sulfur in the rhizosphere [59, 97].

Although SSB reported for increasing sulfur availability and drought tolerance in wheat and maize [245], there are limited information for such benefits in sugarcane. However, in sugarcane, direct evidence connecting microbial sulfur mobilization to improved sulfur nutrition, physiological performance, or drought tolerance is currently limited. Though direct evidence of sulfur-solubilizing bacteria (SSB) in sugarcane is limited, PGPR with known phosphate and potassium solubilizing capacities may also support sulfur mobilization. Its deficiency impairs growth, reduces cane quality, and lowers sucrose yield. SSBs enhance sulfur uptake, stimulate root development, and improve overall plant vigor, reducing the dependency on chemical sulfur fertilizers [118].

9.1 Indirect mechanisms of PGPR enhancing sugarcane drought resilience

Indirect mechanisms of PGPR do not directly promote plant growth; instead, they play a critical role in enhancing plant resilience under drought stress by modulating the rhizosphere environment. These mechanisms alleviate drought-induced ethylene stress, suppress phytopathogens, and strengthen plant defense responses. Major indirect pathways include biological control of pathogens, ACC deaminase activity, and the production of siderophores, ammonia, and hydrogen cyanide (Table 2).

- Biocontrol of Pathogens

Drought stress leads to changes in plant physiology and, therefore, in plant vigor and rhizosphere microbial community, impacting susceptibility to disease and the production of pathogens. Water deficit can increase plant susceptibility to disease by weakening structural barriers, reducing the synthesis of defense-related metabolites, and redirecting carbon resources toward osmoprotection rather than immune responses [157, 171]. Under moderate drought conditions, PGPR can function as biological buffers by suppressing phytopathogens through multiple mechanisms, including antibiotic production, siderophore-mediated iron competition, secretion of lytic enzymes, emission of

Table 2 Indirect mechanisms and their associated PGPR in drought stress mitigation

Mechanism	PGPR examples	Sugarcane validation	Impact	Experimental site	Plant response	References
Biocontrol of Pathogens	<i>P.fluorescens</i> , <i>B. subtilis</i>	Very limited	Moderate	Pot/Field	Reduced disease pressure	[234]
Siderophore Production	<i>P. putida</i> , <i>Bacillus</i> spp.	Limited	Moderate	Pot/greenhouse	Improved Fe acquisition	[170]
Ammonia	(A) <i>brasilense</i> , (B) <i>subtilis</i> , <i>P. fluorescens</i> , <i>Rhizobium</i> spp.	Indirect evidence	Weak	Laboratory/pot	Growth promotion	[8]
HCN Production	<i>P.fluorescens</i> , <i>P. putida</i> , <i>P. aeruginosa</i> Burkholderia cepacia,	Indirect evidence	Weak	Laboratory/pot	Growth promotion	[8, 19, 169]
ACC Deaminase Production	<i>A. brasilense</i> , <i>Pseudomonas</i> spp.	Limited	Moderate	Pot studies	Improved root growth and stress tolerance	[92]

volatile organic compounds, and rhizosphere niche exclusion [56, 229]. During severe or extended drought, reduced microbial activity might lead to decreased antibiotics, lytic enzymes, siderophores, and other antimicrobial compounds and thus reduce the efficiency of biological control [38, 73]. The antagonistic activity of PGPR against pathogens like *Fusarium spp.* and *Rhizoctonia spp.*, the primary complication that can occur in sugarcane, including red rot caused by *Colletotrichum falcatum* and *Ceratocystis paradoxa*-related pineapple disease [106]. Nevertheless, very few studies have assessed whether PGPR-mediated inhibition of these economically critical sugarcane pathogens plays an important role in enhancing field-adapted drought resistance [40, 41].

A key mechanism underlying PGPR-mediated disease suppression is induced systemic resistance (ISR), which primes plant defense systems for faster and stronger activation even under water-limited conditions [161]. PGPR can cause ISR and therefore increase plant coping capacity against both biotic and abiotic stresses through regulation of defense system signaling pathways. Nonetheless, most of the available mechanistic knowledge for ISR is derived from model plants, such as *Arabidopsis thaliana* and other non-sugarcane plants [236]. While activation of jasmonic acid, ethylene, and salicylic acid-dependent defense pathways via PGPR-mediated responses have been described in a few plant species, there is little molecular evidence of a similar effect in drought-stressed sugarcane [240]. Therefore, the hypothesis that ISR can promote PGPR-induced drought tolerance of sugarcane is lacking, which should be further validated with transcriptomic, proteomic, and gene expression studies.

Several PGPR genera, including *Pseudomonas*, *Bacillus*, and *Streptomyces*, produce antimicrobial compounds capable of suppressing economically important sugarcane pathogens, particularly *C. falcatum* and, to a lesser extent, *C. paradoxa*. However, most available studies focus on disease suppression under normal growing conditions, and relatively few have evaluated the stability of these biocontrol interactions under drought stress [168, 228].

Despite these promising findings, pathogen-specific validation of PGPR-mediated biocontrol under field-scale severe drought conditions in sugarcane remains limited. Therefore, future research should integrate moisture-gradient experiments, pathogen virulence assessments, and microbial metabolite profiling to better understand the stability of PGPR-mediated disease suppression under drought stress. Overall, PGPR-based biocontrol should be viewed as a stress-mitigating and risk-buffering mechanism that is most effective under moderate drought conditions, where both plant immune responses and microbial metabolic activity remain functional.

- Siderophore Production

Siderophores are low-molecular-weight, high-affinity iron-chelating compounds that can increase the solubility and mobility of ferric iron (Fe^{3+}) [10, 65, 215]. Several drought-tolerant PGPR genera, including *Pseudomonas*, *Bacillus*, *Azotobacter*, and *Azospirillum*, have been reported to produce siderophores, in conjunction with other adaptive traits such as ACC deaminase activity and phytohormone synthesis, collectively enhancing plant growth, antioxidant capacity, and stress resilience in cereals, legumes, and horticultural crops under moisture deficit conditions [93, 165, 180]. Beyond improving iron nutrition, microbial siderophores also contribute to biocontrol under drought by

competitively restricting iron availability to phytopathogens, suppressing their growth and virulence in the rhizosphere [73, 75].

Drought induced enrichment of bacterial groups like *Rhizobiales* and *Streptomyce-tales* may reflect adaptive restructuring of the sugarcane rhizosphere microbiome, but these insights are largely correlative and cannot be confirmed directly into improved siderophore production or plant iron nutrition [60]. At present, little is known about the association of shifts in the composition of microbial communities with increased iron acquisition by drought-stressed sugarcane. Likewise, although siderophore-bearing PGPR have been found to improve iron uptake and plant performance in some crops, relatively few studies have directly quantified rhizosphere iron dynamics, plant iron accumulation, or leaf iron concentrations in sugarcane following PGPR inoculation under drought conditions. As a result, the association of siderophore-mediated iron mobilization to the drought tolerance of sugarcane is insufficiently characterized. However, the extent to which microbial siderophore production enhances iron uptake by drought-stressed sugarcane remains insufficiently quantified.

- Ammonia and Nitrogen Fixation

Biological nitrogen fixation in sugarcane has been traditionally acknowledged as one of the important sources of nitrogen, especially associated with diazotrophic microorganisms, including *Gluconacetobacter diazotrophicus*, *Azospirillum spp.*, *Herbaspirillum spp.*, and *Burkholderia spp.* However, estimates of the contribution of biologically fixed nitrogen to total plant nitrogen nutrition vary widely depending on sugarcane genotype, soil fertility, environmental conditions, microbial strain, and drought severity. While some investigations have suggested significant nitrogen contributions under favorable conditions, biological nitrogen fixation often supplies only a fraction of total crop nitrogen demand and may decline under severe water deficit [37, 50, 191]. In addition to nitrogen fixation, many PGPR release ammonia as a metabolic by-product, which contributes directly to plant nitrogen nutrition and indirectly suppresses soil-borne pathogens, strengthening plant resilience under drought stress [73, 93]. Microbial nitrogen fixation performance under drought environment is additionally hampered by nitrogenase, the key enzyme responsible for atmospheric nitrogen reduction. Drought-induced alterations in soil aeration, oxygen diffusion and microbial metabolism can also potentially affect nitrogenase activity. Accordingly, while some drought-adapted diazotrophs have protective mechanisms (e.g. osmolyte accumulation, antioxidant defense systems as well as biofilm formation) in place to adapt, biological nitrogen fixation generally declines as drought severity increases [147].

Recent studies further indicate that drought stress reshapes the sugarcane rhizosphere microbiome by enriching drought-tolerant, nitrogen-fixing taxa, highlighting the adaptive significance of microbial nitrogen fixation and ammonia production in sustaining sugarcane productivity under water-limited environments [253]. Collectively, ammonia production and biological nitrogen fixation by PGPR represent key microbial mechanisms that integrate nutrient acquisition, stress tolerance, and plant health, offering a sustainable strategy for improving crop performance under drought-prone agricultural systems. A major knowledge gap is the limited availability of field studies directly quantifying biological nitrogen fixation rates in drought-stressed sugarcane using isotopic techniques and linking these measurements with plant nitrogen status, yield, and

water-use efficiency. Future research should determine how drought intensity, genotype, microbial colonization strategy, and management practices influence the actual contribution of diazotrophs to sugarcane nitrogen nutrition under field conditions.

- Hydrogen Cyanide (HCN) Production

In sugarcane, a crop highly susceptible to both drought stress and soil-borne diseases, PGPR strains with combined drought tolerance and biocontrol traits including HCN production have been reported to enhance plant vigor and stress resilience, while drought-induced shifts in the sugarcane rhizosphere microbiome favor metabolically versatile antagonistic bacteria capable of pathogen suppression [46, 253]. PGPR have typically synthesized HCN in concentrations that are thought to inhibit pathogen activity but not damage plants, although the current evidence for safe concentrations for sugarcane rhizospheres is missing. Furthermore, the modification of root exudation patterns, microbial community composition, soil aeration, and rhizosphere pH in a drought could also potentially affect local HCN levels and bioavailability. Therefore, the tradeoff between the potential beneficial pathogen inhibition, and the possible phytotoxic or ecological effects, need to be determined especially in very severe drought [73, 176, 177].

Several HCN-producing PGPR have shown antagonistic activity against fungal pathogens includes red rot (*Colletotrichum falcatum*), pineapple disease (*Ceratocystis paradoxa*) and smut (*Sporisorium scitamineum*) in laboratory and greenhouse experiments but there is little direct evidence of the role of HCN-mediated biocontrol in drought resilience in sugarcane is inferred rather than experimentally demonstrated [9, 18, 36]. Another significant caveat is that most studies evaluating PGPR for HCN synthesis have used qualitative laboratory assays and have been conducted under optimal growth conditions. In comparison, relatively few studies have measured direct HCN production in drought-stressed rhizospheres or examined how water deficit affects HCN synthesis, diffusion, persistence, and biological activity [176].

- ACC Deaminase Production

In sugarcane, rhizobacteria such as *Pseudomonas* and *Azospirillum spp.* have been reported to enhance root development, biomass accumulation, and drought resilience through ACC deaminase activity, resulting in improved relative water content, chlorophyll retention, and antioxidant enzyme activity under water-limited conditions [168, 202, 204]. ACC (1-aminocyclopropane-1-carboxylate) deaminase plays a pivotal role in mitigating drought-induced ethylene stress in crops. This enzyme cleaves ACC, the immediate precursor of ethylene, into α -ketobutyrate and ammonia, thereby lowering plant ethylene levels and alleviating stress-induced growth inhibition [91, 92, 182]. Through this mechanism, ACC deaminase-producing rhizobacteria enable sustained root growth, improved water uptake, stomatal regulation, and enhanced photosynthetic efficiency, ultimately contributing to improved tolerance to moisture-deficit conditions [63, 91, 229]. ACC deaminase-mediated mitigation of ethylene stress is expected to be most effective in mild- to moderate-drought than drastic water deficiency, although this hypothesis has not been rigorously tested in sugarcane field environments. A notable drawback of the ACC deaminase mechanism is its reliance on active microbial metabolism and sustained rhizosphere colonization.

ACC deaminase activity is highly strain-specific and varies considerably among plant growth-promoting rhizobacteria [139]. The expression of the *acdS* gene encoding ACC deaminase is strongly influenced by environmental conditions, substrate availability, and the type of carbon source present in the rhizosphere [92]. As a result, not all ACC deaminase-producing isolates provide equivalent benefits in improving plant drought tolerance. Since, ACC deaminase activity also depends on active bacterial metabolism, prolonged or extreme desiccation may limit enzyme production and reduce rhizosphere colonization efficiency [143, 191]. Therefore, the effectiveness of ACC deaminase-mediated regulation of plant ethylene levels may decrease under extreme water limitation. These limitations highlight the importance of selecting drought-resilient and metabolically stable bacterial strains for successful field applications.

However, ACC deaminase is regarded as one of the most promising PGPR traits for drought mitigation, several important knowledge gaps still exist in sugarcane. Limited information has been reported on ACC deaminase activity ranges among sugarcane-associated bacterial isolates, soil moisture thresholds at which it can remain active, and the performance of various strains in different field settings. Further research to include the microbial functional assays into a direct assessment of the ethylene dynamics in different drought intensities will be needed to show if and how ACC deaminase impacts the structural resilience of sugarcane under drought conditions.

A key drawback of studies on the indirect PGPR-mediated drought tolerance mechanisms is the absence of uniform quantitative techniques between studies. Attributes including ACC deaminase activity, siderophore production, ammonia production, and HCN synthesis are usually reported qualitatively or measured using different methodologies, thus rendering cross-strain comparisons difficult. Furthermore, many microorganisms reported to have these traits have been evaluated in model crops or laboratory systems rather than in sugarcane under field drought conditions. Consequently, the relative contribution of such individual indirect mechanisms in sugarcane to drought adaptation remains inadequately quantified and should be interpreted with caution.

10 Single-strain vs consortium-based PGPR approaches

Distinguishing between single-strain inoculants and microbial consortia is an important issue for the evaluation of PGPR-mediated drought tolerance. Single-strain inoculants make sense for mechanistic studies because they permit determination of individual microbe characteristics, ACC deaminase activity, phytohormone production, exopolysaccharide synthesis, nutrient solubilization, and biological nitrogen fixation [187, 250]. However, their efficacy could be restricted in some field conditions, where a number of stress-responsive functions require simultaneous application. Microbial consortia is a combination of two or more compatible microorganisms with complementary functional groups, including but not limited to nutrient mobilization, phytohormone production, nitrogen fixation, and antioxidant defense [30, 146]. These actions can support a range of complementary tasks within microbial communities, promoting plant growth and drought tolerance. However, previous studies have demonstrated that consortium-based inoculants perform better in the field than single strains of the same plant, resulting in higher quality. These positive interactions can differ because microbial relations may be weak, neutral, or antagonistic, and such interactions can inhibit the efficacy of microbial assays due to competition for nutrients, ecological niches, or root colonization

sites. It is probable that variation in biofertilizer performance could also depend on variations in microbial composition, formulation quality, storage conditions, and post-application persistence [78]. In sugarcane, direct comparison between single-strain and consortium-based inoculants is very limited. For example, *Bacillus megaterium* strains JTB1 and MT22 and their consortium improved sugarcane growth, root architecture, antioxidant activity, and photosynthetic gene expression under drought, but the consortium did not consistently outperform the individual strains across all measured traits [41]. These approaches should further be in the same environmental conditions to see if microbial consortia have a greater resilience against drought in different sugarcane genotypes, soils, and production systems.

11 Integration, trade-offs, and context dependency of PGPR-mediated drought responses

PGPR-mediated drought tolerance in sugarcane arises from the interaction of multiple microbial traits, plant physiological processes, and environmental conditions, not a single microbial function. Functions such as ACC deaminase activity, indole-3-acetic acid (IAA) production, exopolysaccharide (EPS) synthesis, phosphate solubilization, biological nitrogen fixation, and siderophore production together influence root development, water and nutrient uptake, hormonal regulation, osmotic adjustment, and antioxidant metabolism [40, 197]. However, the co-occurrence of several beneficial traits does not necessarily imply a synergistic effect of these traits. An additive response occurs when the combined effect of two or more traits is approximately equivalent to the sum of their individual contributions, whereas a synergistic response occurs when their interaction produces an effect greater than that expected from the individual effects.

An interaction that is potentially complementary is IAA-driven root development and ACC deaminase-mediated regulation of stress-induced ethylene. IAA stimulates primary and lateral root growth, while ACC deaminase reduces stress-associated ethylene that would otherwise limit root elongation [147, 197]. However, this relationship is not necessarily synergistic. Too much auxin can increase biomass and water consumption, which could be counterproductive in extreme drought, and the optimal balance is probably determined by the trade-off between stimulation of growth and water saving. However, there is no factorial study that isolates this specific interaction in sugarcane, and hence, this is a testable hypothesis and not an established impact.

A similar caveat applies to nutrient-mobilizing traits. Mineral solubilization, nitrogen fixation, and siderophore-mediated iron acquisition can each independently support photosynthesis and antioxidant metabolism under drought, and their combined presence may simply be additive [40]. Confirming true synergy requires evidence that combined effects exceed the sum of individual ones, currently insufficient for sugarcane. There may also be physiological trade-offs: producing siderophores, EPS, hormones, or enzymes all draw on shared microbial resources, so investment in one function could limit another depending on carbon/nutrient status, growth phase, and environmental stress. A proposed trade-off between siderophore production and ACC deaminase activity is mechanistically plausible but not yet experimentally confirmed through direct measurement of microbial resource allocation.

Field data point to the need for restraint in assuming consortia to be synergistic by default. *Bacillus megaterium* strains JTB1 and MT22, used individually and as a

consortium, improved sugarcane root growth, antioxidant, and molecular responses in the face of drought, but the consortium did not always outperform the individual strains [41]. This study demonstrates that combining strains can result in trait-specific, neutral or additive effects, rather than universal synergy, depending on strain compatibility, competition, colonization sites, persistence, and host genotype thus, consortia should be benchmarked against their individual components under the same conditions.

Trade-offs also occur at the plant level increasing root growth improves water/nutrient uptake, but at a carbon cost, and gains may plateau with increasing drought severity. So one improved characteristic doesn't necessarily translate to whole-plant drought resistance it's really about whether the combined microbial effects maintain plant water status, photosynthesis, biomass, and eventually cane/sugar yield. The efficacy of PGPR is not an intrinsic characteristic of a strain but depends on the prevailing conditions, including plant genotype, soil properties, drought intensity, temperature, nutritional status, formulation, and timing of inoculation. Future studies should employ factorial, multi-strain designs to simultaneously quantify colonization, physiology, and yield, confirmed across various seasons and locales, rather than assuming synergy from the simple existence of numerous favourable features.

12 Biocontrol, drought stress, and disease interactions

Biocontrol may indirectly increase the drought resilience of sugarcane by relieving pathogen stress, but this should be decoupled from the direct physiological drought tolerance mechanisms of PGPR. Water shortages can aggravate red rot (*Colletotrichum falcatum*), and different sugarcane varieties respond differently to it. The rhizosphere bacteria include species of *Bacillus*, *Pseudomonas*, *Gluconacetobacter*, *Ochrobactrum*, and *Acinetobacter*. In vitro, selected strains have shown antagonism to *C. falcatum* and, in greenhouse tests, have reduced disease severity and promoted development [106, 228, 240]. However, these strains need to be validated in natural habitats to be deemed reliable biocontrol agents.

Importantly, biocontrol efficacy under adequate moisture cannot be assumed to hold under severe drought. Water deficit alters microbial growth, root exudation, rhizosphere connectivity, and pathogen–host dynamics, and direct evidence quantifying biocontrol performance loss under drought in sugarcane remains limited [35, 248]. Most biocontrol studies to date were conducted under lab, greenhouse, or well-watered conditions rather than combined drought $\times$ pathogen stress [26, 49, 235].

Biocontrol should therefore be treated as a potentially complementary, not demonstrated, component of PGPR-mediated drought tolerance in sugarcane. Future work should use factorial designs (water regime $\times$ pathogen challenge $\times$ microbial inoculation) measuring pathogen development, microbial survival/colonization, plant water status, disease severity, biomass, and final yield, to determine whether disease suppression persists under severe drought and whether effects are additive, synergistic, neutral, or antagonistic.

13 Temporal dynamics of PGPR responses during drought development and recovery

The response of plant growth-promoting rhizobacteria (PGPR) to drought is now understood as a dynamic, multi-phase process rather than a static condition, occurring over discrete stages of stress onset, progression, peak severity, and post-rewatering recovery. In the early phase of water deficit, PGPR rapidly alter host ethylene signaling through ACC deaminase activity and initiate an upregulation of exopolysaccharide production, processes that buffer the rhizosphere microenvironment before the accrual of more severe physiological damage. The timing of inoculation relative to stress onset also appears to shape efficacy: in sugarcane cv. SP70-1143, *Gluconacetobacter diazotrophicus* PAL5 applied seven days prior to the imposition of water deficit provided one of the clearest demonstrations of pre-emptive PGPR conditioning ahead of stress [223]. As stress intensifies, this response transitions toward osmolyte biosynthesis, phytohormone-mediated root architectural remodeling, and priming of host antioxidant enzyme systems, alongside compositional shifts in the broader rhizosphere community that favor desiccation-tolerant, spore and biofilm-forming taxa.

This temporal sensitivity is further modulated by irrigation regime and host genotype. A study by May, Moreira [132] reported that *B. aryabhattai* produced more pronounced growth-promoting effects at 7- and 14-day irrigation intervals, with the magnitude of the response varying across genotypes. Similarly, a field trial by Almeida, Santos [15], in which pre-sprouted seedlings were inoculated with *B. licheniformis* and *B. subtilis*, found that inoculation most strongly enhanced crop growth rate under the most restricted water regimes, with these benefits becoming increasingly evident as plant establishment progressed. Comprehensive reviews of PGPR-mediated drought tolerance have consolidated evidence for several key mechanisms, including nitrogen fixation, phosphate solubilization, hormone production, and antioxidant regulation. These reviews also indicate that PGPR are generally more effective at enhancing photosynthesis and limiting oxidative damage under drought conditions than under non-stress conditions, underscoring the stress-contingent nature of these benefits.

Critically, the recovery phase following rewatering is emerging as a mechanistically distinct rather than merely reversive stage. Research by Santos, Edward [184] on rice root microbiomes has shown that drought-stressed root communities are developmentally delayed relative to well-watered communities and that the duration of prior drought exposure shapes the subsequent trajectory and rate of microbiome recovery. Parallel high-resolution metabolomic work on rewatered sugar beet plants has similarly demonstrated that shoot and root tissues follow divergent recovery kinetics, with a transient normalization of most metabolites within roughly eight days followed by a secondary accumulation of amino acids in leaves that may represent a beneficial stress imprint for future drought events [233]. Together, these findings suggest that PGPR do not simply retreat to a pre-stress baseline upon rewatering but instead retain functional and compositional legacies potentially underlying priming or “stress memory” effects that merit dedicated time-course, multi-omics investigation rather than the endpoint sampling that has dominated the field to date.

14 Technological innovations in the use of PGPR for sustainable agriculture

Recent developments in microbial biotechnology offered novel avenues for enhancing the performance and reliability of PGPR (Plant Growth-Promoting Rhizobacteria) under drought. Instead of targeting genetic engineering in a host plant, increasing attention in this context is turned to finding, selecting, and modifying microbial characteristics that are directly related to stress relief. Genomics, transcriptomics, proteomics, and metabolomics approaches are combining to identify genes implicated in ACC deaminase activity (*acdS*), biological nitrogen fixation (*nif* genes), siderophore biosynthesis, osmoprotectant production, exopolysaccharide synthesis, and rhizosphere colonization [4, 17, 238]. These technologies allow the identification of drought-resilient strains with better functional stability under various environmental conditions. Marker-free genome-editing methods and targeted genetic engineering have also been investigated for improving desirable microbial traits; however, most of these methods have not been verified in field sugarcane systems and are currently at the experimental phase [85, 202]. Innovations in the inoculant production process, besides strain enhancement, are also improving the survival and efficacy of PGPR under water-limited circumstances. Synthetic microbial communities with co-evolved functional traits of complementary nature, including ACC deaminase production, nutrient mobilization, and osmotic stress tolerance, may hold a better stability than single-strain inoculants in any range of heterogeneous environmental conditions [201]. Similarly, microbial encapsulation, polymer-based carriers, and nano-enabled delivery systems improve shelf life, rhizosphere establishment, and protection against environmental stresses [78, 196]. Despite these enhancements, field validation in sugarcane is scarce, and the performance of engineered or highly selective PGPR strains under extreme drought conditions, different soil types, or various sugarcane genotypes needs to be further explored prior to large-scale commercial applications of PGPR compounds [41, 52].

15 Commercialization and adoption of PGPR technologies in sugarcane

Growing interest in sustainable agriculture has driven the commercialisation of many microbial inoculants using plant growth-promoting rhizobacteria (PGPR) including *Bacillus*, *Azospirillum*, *Pseudomonas*, *Rhizobium*, and *Gluconacetobacter* genera. Several of these microorganisms are sold worldwide as biofertilizers, biostimulants, or products that help plant growth, and have exhibited beneficial effects on nutrient acquisition, root formation, and crop production following a variety of environmental conditions [24, 64, 66]. For sugarcane, the microbial products that contain nitrogen-fixing bacteria as well as growth-promoting bacteria such as *Gluconacetobacter diazotrophicus*, *Herbaspirillum* spp., *Azospirillum* spp., and *Bacillus* spp. have demonstrated potential for enhancing crop growth and increasing efficiency in nutrient-use [5, 67, 167]. Even with encouraging experiment results, only a few commercial products have been developed and validated specifically for drought mitigation in sugarcane [150] (Table 3). One important factor is the gap between the efficiency of greenhouses and work performance in field work. Such environmental variability, soil moisture and temperature variations, soil physicochemical characteristics, and direct interaction with indigenous microbial communities can markedly affect inoculant survival and efficacy under field conditions [114, 149].

In turn, strains that are consistently effective under controlled conditions have the potential to show varying responses for various locations, seasons, and production

Table 3 Commercially utilized and experimentally validated PGPR genera, with emphasis on drought-related applicability

PGPR genus/species	Commercial formulation type	Key mechanism(s) of action	Documented Benefits in sugarcane	Typical application method	References
<i>Bacillus</i> spp. (e.g., <i>B. subtilis</i> , <i>B. amyloliquefaciens</i> , <i>B. megaterium</i>)	Biofertilizer/biostimulant; spore-based liquid or carrier formulations	Phosphate and potassium solubilization; ACC-deaminase activity; production of exopolysaccharides (EPS) and osmolytes; induced systemic resistance	Improved nutrient acquisition, root proliferation, and stress tolerance under variable field conditions	Seed/sett treatment, soil drenching, or furrow application at planting	[66, 150]
<i>Azospirillum</i> spp. (e.g., <i>A. brasilense</i> , <i>A. lipoferum</i>)	Nitrogen-fixing biofertilizer, liquid/carrier-based inoculant	Associative N ₂ fixation; phytohormone (IAA, gibberellin) production; root hair proliferation	Enhanced root growth and nitrogen-use efficiency, allowing partial reduction of synthetic N inputs Tiller count, stalk diameter	Sett/seedling inoculation or soil application	[5, 167]
<i>Herbaspirillum</i> spp. (e.g., <i>H. seropedicae</i>)	Endophytic N-fixing inoculant, often used in consortia with <i>Gluconacetobacter</i>	Endophytic colonization; nitrogen fixation; phytohormone modulation	Contributes to biological nitrogen fixation in sugarcane in combination with other diazotrophs	Sett/seedling inoculation	[192]
<i>Pseudomonas</i> spp. (e.g., <i>P. fluorescens</i> , <i>P. putida</i>)	Biocontrol agent and growth promoter; wettable powder or liquid formulations	Siderophore production; phytohormone synthesis; antagonism against soil-borne pathogens; ACC-deaminase activity	Growth promotion combined with suppression of soil-borne pathogens, indirectly supporting plant vigor under stress	Root-zone application, seed coating	[66]
<i>G. diazotrophicus</i>	Sugarcane-specific endophytic inoculant	Endophytic nitrogen fixation; gluconic acid-mediated phosphate solubilization; phytohormone production	Well-documented endophytic colonization of sugarcane tissues; contributes to the nitrogen economy of the crop	Sett soaking, in-furrow application	[5, 192]
<i>Rhizobium</i> spp.	Biofertilizer	Nitrogen fixation (associative, non-nodular in non-legumes); phytohormone production; siderophore synthesis	Limited but emerging evidence of growth-promoting associative effects in sugarcane	Seed/sett treatment	[66]

systems. Further challenges are short shelf life, formulation instability, storage, and difficulty to achieve microbial survival during the preparation and application. In addition, achieving commercialization will necessitate proving that agronomic benefits are stable at various agronomic systems across a range of environments and adhering to certain regulations and the economic viability for farmers to make the product commercially viable. Robust microbial consortia, better formulations of carriers, precision delivery technologies, and large-scale multi-location validation trials will enhance future improvements. Integration of microbial inoculants with drought-tolerant cultivars, better management of irrigation, and climate-smart agricultural practices can complement the reliability and deployment of PGPR-based technology to sugarcane production systems.

16 Recent sugarcane-specific evidence from field and water-deficit Studies

The current data for sugarcane-specific PGPR is heterogeneous in terms of setting, strain, genotype, and results, both in field and controlled-environment research, and should not be considered equally validated. A recent field trial in Brazil has shown that sugarcane inoculated with *Bacillus licheniformis* and *B. subtilis* showed better germination count, water-use efficiency, and improved root/shoot growth under different degrees of water stress (0%, 33%, 66%, and 100%) [15]. Also, Carnietto, Santos [44] have recently reported that the effects of inoculating *Bacillus subtilis* FMCH002 and *Bacillus licheniformis* FMCH001 under water scarcity were dependent on the soil type. Thus, the efficiency of PGPR is not only dependent on the bacterial strain but also on soil conditions. However, the work does not give any direct confirmation in the field, since it was conducted under controlled laboratory circumstances.

A pot experiment with *Bacillus megaterium* strains JTB1 and MT22 (alone and as a consortium) under 100/50/25% field capacity revealed improved root development, biomass, antioxidant responses, and photosynthetic properties. But the individual strains performed usually the same as the consortium, which indicates that strain combinations are not necessarily better than inoculating with a single strain [41]. Differential responses of sugarcane genotypes to microbial inoculation have been evidenced recently. In recent studies, growth, biomass, and nitrogen content were measured using a diazotrophic consortium of *Gluconacetobacter diazotrophicus* and other beneficial bacteria [68, 70, 191]. But there is scant concrete evidence that these inoculants may alleviate drought stress in field conditions. Thus, positive effects of *G. diazotrophicus* on nitrogen fixation or plant development should not be taken as proof of drought protection. In general, the recent published work added to the sugarcane-specific data base, but consistent benefits for mature-crop cane and sugar yield during drought remain inadequately proven. The variation among genotypes of sugarcane, soil types, bacterial strains, and water regimes further suggests that the effectiveness of PGPR should be evaluated as a genotype $\times$ microbial inoculant $\times$ environmental interaction rather than as a universal attribute of a particular strain.

17 Current evidence and remaining uncertainties

PGPR have also been reported to exert positive effects on sugarcane growth and drought-induced physiological responses; however, the magnitude and consistency of these effects vary considerably among response variables and experimental systems.

Several greenhouse and pot studies have reported improvements in nutrient uptake, antioxidant activity, osmotic adjustment, root development, and water-use efficiency following PGPR inoculation [15, 41, 86, 193].

Broader meta-analyses and systematic reviews further support the potential of PGPR to increase plant tolerance to drought stress, but responses are generally not standardized and depend on species, drought severity, bacterial qualities and experimental design [114, 255]. As a result, controlled experiments with good results should not be assumed to be replicable for all sugarcane-producing environments. A considerable area of uncertainty in present research is the lack of multi-site, multi-season field trials that examine inoculation performance in naturally fluctuating drought regimes, soil, climate and management conditions. Field efficiency might be affected by the severity of drought, soil physicochemical properties, temperature, sugarcane genotype, inoculation method, formulation quality and, to some extent the interactions at the field level with indigenous microbial communities [52, 149]. As a result, the different effects of PGPR strains, which have consistent performance in greenhouse or pot conditions, may vary by regions, seasons and production systems. Consequently, the current evidence indicates that PGPR has potential as a well-studied crop control technique for enhancing drought-resistant sugarcane yields, but issues regarding reproducibility, effect size, long-lasting effect, yield stability, and commercial scale application should be addressed as well. Future studies should report the success of the inoculants, and the failure or inconsistency of the inoculants per se in producing agronomic values under field conditions [78, 83].

18 Limitations and inconsistent responses of PGPR under drought conditions

The PGPR-mediated drought tolerance in sugarcane is not universal but conditional and is significantly affected by the genotype, microbial identity, inoculation approach, nutritional status, and environment. Improvement of physiological/biochemical features does not always lead to yield improvement. PGPR inoculation increased nutrient acquisition, technological quality, and yield, but only under intermediate phosphorus-fertilization levels, with no positive effect at zero P or at the highest P_2O_5 rate, suggesting that the response is dependent on the surrounding nutrient environment rather than an inherent inoculant property [169]. Likewise, *Bacillus megaterium* strains JTB1 and MT22, alone or as a consortium, under moderate/severe drought enhanced root growth and antioxidant and molecular traits, but the consortium did not perform better than single strains [39], indicating that mixing strains does not automatically improve agronomic response. Strong laboratory/greenhouse performance also should not be read as evidence of field reliability. In the field, PGPR face competition from native microbes, limited root niches, and variable soil moisture, temperature, and physicochemical conditions; moderate-to-severe drought further limits microbial activity, colonization, and metabolite production [60, 123, 127]. Since inoculant persistence and colonization are rarely measured directly in drought-stressed sugarcane studies, it's often unclear whether a null result reflects true inoculant failure or simply poor establishment.

Competition with the native rhizosphere microbiome adds further variability: introduced strains compete for nutrients, carbon, and spatial niches with adapted native microbes, and drought-altered root exudation can favor natives over introduced strains. Genotype, root architecture, soil texture, organic matter, pH, nutrient availability, drought intensity, temperature, and management practices together drive substantial

spatial and temporal variation in PGPR performance [48, 213], explaining why controlled-condition successes often shrink, flatten, or disappear under field conditions. Publication bias is a further limitation: positive, statistically significant results are published and cited more often than null or mixed outcomes, likely inflating the apparent success rate of PGPR, a particular concern given how few field (versus controlled-environment) drought studies exist in sugarcane. Neutral or unsuccessful outcomes should be treated as integral evidence.

Overall, current evidence supports a conditional response model in which absence of yield improvement should not be read as complete biological failure when intermediate traits (root development, nutrient acquisition, antioxidant activity, water-use efficiency) improve nor should improvement in those traits be assumed to guarantee yield success. Future studies should jointly measure microbial survival/colonization, physiological responses, and final yield across multiple environments and seasons to separate genuine efficacy from context-dependent artifacts.

19 Challenges and future perspectives

There are several studies describing the feasibility and extent of PGPR applications in agricultural applications, which raises translation problems from controlled studies into real-world applications. Research showing an impact on drought tolerance is strong in the literature, including studies on PGPR, yet most of the available research is conducted using greenhouses, growth chambers, and pot experiments in controlled environmental conditions. These studies have advanced our knowledge of microbial responses to drought, including phytohormone production, nutrient mobilization, ACC deaminase activity, osmotic adjustment, and antioxidant regulation. Controlled conditions are not able to fully recapitulate the complex and variable conditions of a field environment. Under field conditions, different interacting factors such as temperature swings, rain variability, soil heterogeneity, irrigation, nutrient availability, and competition with indigenous species may affect PGPR-introduced strains with respect to their survival, colonization, and functional response. As such, microbial inoculants that exert strong growth-facilitating effects under controlled environments might perform heterogeneously in agricultural habitats [19, 55]. Controlled experiments often report 20–60% improvements in plant growth parameters following PGPR inoculation, whereas field trials typically show 5–25% gains or inconsistent results due to environmental variability. Multi-location field evaluations frequently report variable responses across sites and growing seasons, highlighting the difficulty of translating laboratory efficacy into consistent field performance [22, 154]. This gap between laboratory success and field reliability remains a key barrier to their widespread adoption.

Competition with indigenous soil microorganisms further limits the establishment and persistence of introduced PGPR strains. Native microbes, being well adapted to local conditions, often outcompete inoculated strains for nutrients and ecological niches. Additionally, drought-induced changes in root exudation patterns can disrupt microbial recruitment and reduce PGPR efficiency [19, 213]. Crop genotype also influences outcomes, as variations in root architecture, exudate composition, and plant immunity affect microbial colonization, making it difficult to develop universally effective inoculants for diverse sugarcane cultivars [154].

Soil physicochemical properties, including pH, nutrient status, and organic matter content, along with environmental factors such as extreme temperature and irregular rainfall, further contribute to variability in PGPR performance [32]. These challenges are particularly significant in drought-prone regions, where stress conditions are more severe. Moreover, limited understanding of complex plant-microbe-soil interactions under drought conditions restricts the ability to optimize PGPR functionality at the field level.

Commercialization of PGPR-based products also presents challenges, including the need for cost-effective large-scale production, improved formulation technologies, and enhanced shelf life. Despite their environmental benefits, these factors can increase production costs and limit adoption by farmers. Future prospects lie in integrating PGPR with advanced agricultural practices and emerging technologies. The development of robust microbial consortia, improved bioformulations, and better delivery systems can enhance microbial survival and effectiveness. In addition, combining PGPR applications with precision agriculture tools such as soil health monitoring, climate-smart irrigation, and digital decision-support systems can optimize their field performance. In conclusion, while PGPR hold significant potential for improving drought resilience and sustainability in sugarcane cultivation, overcoming challenges related to field variability, microbial establishment, formulation, and large-scale application are essential. Continued research, technological advancements, and supportive policy measures will be critical to fully realize the potential of PGPR in climate-resilient agriculture.

In the context of sugarcane, field-based drought studies report less evidence than greenhouse studies. Moreover, the vast majority of field evaluations published to date have been performed at individual sites or a single growing season, limiting the ability to evaluate the long-term stability as well as reproducibility of inoculant performance across different environmental conditions. Limited studies have investigated cane yield, sucrose accumulation, water-use efficiency, and microbial persistence under natural drought stress across single sites and multiple seasons. The discrepancy between experimental and field effectiveness is one of the main obstacles for scaling PGPR technology up. The successful field application does not stop at the development of valuable microbial strains but also includes formulation stability, shelf life of inoculants, rhizosphere competitiveness, environmental adaptation, and general performance compatibility in different production systems. Thus, the vast empirical evidence indicates the potential for PGPR to be applied to sustainable drought management but large scale potential needs to be interpreted with wariness and skepticism until well-researched in a wide range of agroecological sites and growing periods [19, 33].

Economic feasibility is a major obstacle to the extensive use of PGPR technologies in sugarcane cultivation. Sugarcane is generally cultivated as a low-margin, high-volume crop, particularly in developing countries, where the cost of production is a significant determinant of management choices. These PGPR inoculants increase nutrient-use efficiency and drought resistance potential, but their economic advantages must always surpass the costs of inoculant production, formulation, transportation, storage, and application in order for inoculation to be an economic benefit [149, 167]. Currently, a relatively limited number of studies have performed full cost-benefit analyses on PGPR applications under drought-driven sugarcane production systems, which may make the economic value of such methods unable to be accurately measured in a commercial

context. Another significant challenge is that of inoculant formulation and storage stability [14, 64].

Most PGPR strains have poor viability during long storage, especially in tropical and subtropical areas with high humidity and temperature conditions. Poor microbial viability during storage can severely deteriorate field performance and cause variations in the effects. Despite carrier-based formulations, microbial encapsulation technologies and polymer-protected bioformulations have all increased shelf life, maintaining sufficient cell viability during transportation and on-farm storage remains a major challenge, especially in areas of the world with limited cold-chain infrastructure [78, 113]. Farmer adoption is another bottleneck. The introduction of PGPR technologies may be successful not only based on biological efficiency but also depends upon farmer awareness, availability of high-quality products, knowledge and confidence in efficacy, and compatibility with existing systems. In the majority of regions where sugarcane is planted, it can be believed that microbial inoculants are potentially high-risk inputs in many cases due to their short-term effects being less predictable and less efficient than traditional fertilizers [149, 153]. Variability in product efficacy, lack of extension support, and immature distribution networks could also limit adoption. Thus, better farmer education, better quality control standards, and reliable supply chains can all be keys to successful PGPR implementation [64, 66].

Interactions between microbial inoculants and traditional agricultural inputs are also underdeveloped. Fertilizer, fungicides, insecticides, herbicides, and seed-treatment chemicals could influence PGPR survival, colonization, and function. Some agrochemicals are antimicrobial and may inhibit establishment of microbes or change rhizosphere community composition, thus reducing inoculant efficacy [185]. In contrast, some integrated nutrient management systems might improve the performance of PGPR by generating complementary resources. It is thus important to further investigate PGPR compatibility with widely used sugarcane production inputs under field conditions.

Another fundamental challenge is scaling up the experiment from laboratory studies to commercial production. Microbial strains often show reduced efficacy under field conditions compared to greenhouse conditions because of the variability of environmental conditions, competition with native microbial communities, and heterogeneity of soil characteristics. In addition, industrial-scale fermentation, downstream processing, formulation development, quality control, and regulatory approval bring additional technical and economic [66, 78]. Genetic stability, functional consistency, and shelf life at large-scale production remain critical for commercial success. Continued research in this field should progress beyond proof-of-concept studies to multi-location and multi-season field trials on heterogeneous soil types, climatic conditions, irrigation regimes, and sugarcane genotypes (Table 4). We must also take a more targeted approach to drought severity thresholds, microbial persistence, inoculation timing, compatibility with agricultural inputs, and economic performance. The integration of PGPR with precision agriculture, climate-smart irrigation, microbial consortia, and novel bioformulation technologies has the potential to improve consistency and adoption. However, the effective use of PGPR in drought-prone sugarcane systems will be a collective responsibility of researchers, industry stakeholders, policymakers, extension agencies, and growers, with particular emphasis on biological efficacy and economic viability. The research in future should focus on the coordinated multi-environment field trials, uniform

Table 4 Direct and contextual evidence of PGP/PGPB effects in sugarcane relevant to drought resilience

Location	PGPR strain/species	Drought severity/water regime/experimental condition	Demonstrated trait/response	Main outcomes measured	Outcome direction/key finding	Reference
Mexico	Systemic biofertilizer: <i>P. fluorescens</i> + <i>A. brasilense</i> + <i>B. subtilis</i>	Contextual evidence — non-drought; Field experiment	No imposed drought; local rainfall/management conditions	Cane yield, stalk traits, *Brix, sucrose/chlorophyll and harvest quality	Provides field evidence for biofertilizer performance, but cannot be considered direct drought evidence	[5]
Lencóis Paulista, São Paulo, Brazil	<i>B. licheniformis</i> FMCH001 (DSM32154) + <i>B. subtilis</i> FMCH002 (DSM32155)	Drought stress imposed on sugarcane seedlings; under field experiment	Improved establishment, root development, nutrient acquisition and water-use efficiency	PSS survival, shoot/root growth, nutrient accumulation, water-use efficiency and stalk multiplication	Under strongest restriction, PSS survival increased by 4.17% and WUE by 185.10%	[15]
Jember, East Java, Indonesia	<i>B. megaterium</i> JTB1 and MT22, individually and as consortium	Drought stress imposed on sugarcane seedlings; under pot/greenhouse experiment 100% FC = normal; 50% FC = moderate drought; 25% FC = severe drought	Combined effects on root architecture, IAA production, ACC-phosphate solubilization, ACC-deaminase activity, antioxidant protection and photosynthetic regulation	<i>Bacillus megaterium</i> JTB1 significantly improved root architecture, biomass/growth, antioxidant response and photosynthetic gene expression. MT22 reduced drought-induced oxidative damage; enhanced expression of antioxidant and photosynthetic genes including CAT, APX, RBC-L, PEPC and SPS.	Root development markedly improved under severe drought (25% FC)	[41]
Brazil	<i>B. subtilis</i> FMCH002 + <i>B. licheniformis</i> FMCH001	Drought stress imposed on sugarcane seedlings; under field experiment	Improved physiological and biochemical regulation, biomass production and stress responses; effectiveness modified by soil texture Root development, soil microbial functioning, nutrient acquisition, photosynthetic performance and water-use efficiency; improved recovery after rewetting	Physiological/biochemical responses, root/shoot biomass and soil-texture-dependent responses. Improved PSII efficiency, stomatal conductance, carboxylation efficiency, leaf water potential and relative water content under relevant conditions. Inoculation increased root length, root surface area and diameter under severe water restriction. PGPB increased root dry matter and shoot dry matter in sandy soil.	Demonstrates strong environmental / context dependence; controlled soil experiment rather than field-scale production. Soil texture substantially modified the response. Root dry mass + 17.26% and shoot dry mass + 15.45% overall; stronger gains in sandy soil	[44]
Subtropical India	<i>B. subtilis</i> BSSC11 + <i>B. megaterium</i> BMSE7	Drought stress imposed on sugarcane seedlings; drought vs. irrigated conditions under field condition	Growth promotion, nutrient acquisition, antioxidant defence and osmotic adjustment under drought	Early sprouting, shoot/root growth and biomass, nutrient accumulation, antioxidant enzymes, proline	Positive effects on growth- and tolerance-related traits under drought	[48]

Table 4 (continued)

Location	PGPR strain/species	Drought severity/water regime/experimental condition	Demonstrated trait/response	Main outcomes measured	Outcome direction/key finding	Reference
Brazil	five diazotrophic bacteria, <i>G. diazotrophicus</i> , <i>Herbaspirillum seropedicae</i> , <i>H. rubrisubalbicans</i> , <i>Nitrospillum amazonense</i> , and <i>Paraburkholderia tropica</i>	Drought stress imposed on sugarcane seedlings; under field experiment	Reported severe drought/low-precipitation conditions	Stalk yield, Brix, polarizable sugar and leaf N	Drought-related field evaluation	[192]
Botucatu, São Paulo, Brazil	<i>B. subtilis</i>	Drought stress imposed on sugarcane seedlings; under pot/greenhouse experiment	Improved nutrient acquisition, physiological performance, photosynthetic activity and antioxidant/stress responses	Nutrient uptake, photosynthetic pigments, gas exchange, antioxidant metabolism, root and stalk growth	Positive, trait-dependent responses under water deficit	[86]
Australia	Biofertilizer containing organic waste + mineral N, with/without added PGPR	Contextual evidence — non-drought; Field experiment	No defined drought treatment	Biomass, sugar yield, soil mineral N, microbial community composition and N-use efficiency	Provides production and N-use efficiency evidence; drought response was not directly evaluated	[167]
Ilha Solteira, São Paulo, Brazil	<i>(A) brasilense</i> , <i>(B) subtilis</i> and <i>P. fluorescens</i> , individually and combined	Contextual evidence — non-drought; Field experiment	No imposed drought; residual-P experiment	Cane yield, technological quality, nutrient uptake and response to residual P fertilization	Demonstrates effects on crop performance/nutrient response, but not direct drought evidence	[177]
Subtropical India	<i>B. megaterium</i> var. <i>phosphaticum</i>	Contextual evidence — non-drought; Field experiment	No drought treatment; phosphorus-management experiment	Rhizosphere PSB population, available P, tillering, stalk population/weight, cane yield, juice quality and sugar yield	Provides evidence for P mobilization and crop performance, but not direct evidence of drought mitigation	[208]
Tropical India	Bio-intensive microbial/rhizosphere management system	Contextual evidence — non-drought; Field experiment	No defined drought treatment	Soil health, rhizosphere microbial activity and ratoon productivity	Improved soil/microbial and productivity-related parameters; drought mitigation not directly evaluated	[211]

evaluation protocols, and long-term surveillance of microbial establishment and crop response. This type of work will be necessary to translate these promising lab findings into reliable and scalable solutions for drought-resistant sugarcane production systems.

20 Conclusion

Plant growth-promoting rhizobacteria offer a biologically plausible and potentially sustainable approach for supporting sugarcane under water-limited conditions. The literature reviewed in this article indicates that PGPR can influence several processes relevant to drought adaptation, including root development, nutrient mobilization, biological nitrogen fixation, phytohormonal regulation, ACC deaminase activity, osmotic adjustment, antioxidant defense, rhizosphere stabilization, and pathogen suppression. However, the strength of evidence is not uniform across these mechanisms. The most consistent evidence concerns improvements in plant growth, nutrient acquisition, root traits, and selected physiological responses, whereas direct evidence linking individual microbial mechanisms to improved cane and sugar yield under naturally occurring drought remains comparatively limited.

A central conclusion of this review is that PGPR efficacy should not be regarded as an intrinsic and universally transferable property of a microbial strain. Instead, it emerges from interactions among the inoculant, sugarcane genotype, drought intensity and duration, developmental stage, soil physicochemical characteristics, indigenous microbial communities, irrigation regime, temperature, nutrient status, formulation, and inoculation timing. The rhizosphere therefore represents a critical determinant of inoculant establishment and functional expression. Moderate water stress could show observable benefits from PGPR, but severe and long-term droughts could limit microbial survival, root activity, metabolite production, nutrient diffusion, and colonization, which could potentially lead to a reduction of these effects. In addition, the evidence is a warning against the anticipation that multiple beneficial microbial traits or consortium-based inoculants will necessarily result in synergistic effects. Gains in intermediate traits such as root biomass, chlorophyll, nutrient uptake, antioxidant activity, or water-use efficiency do not necessarily translate into increased cane or sugar yield. Conversely, the absence of a yield response should not necessarily be considered complete biological failure if meaningful improvements are seen in physiological or nutritional traits. Consequently, neutral, inconsistent, and negative results should be incorporated into future evaluations rather than being overlooked.

Three research priorities are particularly important. First, multi-location, multi-season field experiments should quantify microbial survival and colonization together with soil moisture, plant physiology, nutrient acquisition, cane yield, sucrose yield, and water-use efficiency. Second, standardized experimental frameworks should evaluate PGPR responses across drought severity gradients, sugarcane genotypes, soil types, inoculation timings, and drought-recovery phases. Third, development and commercialization should integrate robust strain selection with formulation stability, shelf life, compatibility with fertilizers and agrochemicals, cost–benefit analysis, and farmer-oriented delivery systems. Thus, PGPR should presently be considered a context-dependent component of integrated drought management, with substantial promise but insufficient evidence to support their characterization as a universally reliable solution for drought-resilient sugarcane production.

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